

Journal of Engineering Design

ISSN: 0954-4828 (Print) 1466-1837 (Online) Journal homepage: www.tandfonline.com/journals/cjen20

The power and potential of Behavioural Design: practice, methodology, and ethics

Camilla K. E. Bay Brix Nielsen, Philip Cash & Jaap Daalhuizen

To cite this article: Camilla K. E. Bay Brix Nielsen, Philip Cash & Jaap Daalhuizen (2024) The power and potential of Behavioural Design: practice, methodology, and ethics, Journal of Engineering Design, 35:5, 504-542, DOI: 10.1080/09544828.2024.2322897

To link to this article: https://doi.org/10.1080/09544828.2024.2322897

© 2024 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

Published online: 13 Mar 2024.

Submit your article to this journal

Article views: 6175

View related articles

View Crossmark data

Citing articles: 11 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=cjen20

JOURNAL OF ENGINEERING DESIGN
2024, VOL. 35, NO. 5, 504–542
https://doi.org/10.1080/09544828.2024.2322897

Taylor & Francis
Taylor & Francis Group

OPEN ACCESS Check for updates

The power and potential of Behavioural Design: practice, methodology, and ethics

Camilla K. E. Bay Brix Nielsen [a], Philip Cash [b] and Jaap Daalhuizen [a]

[a]Department of Technology, Management and Economics, DTU Technical University of Denmark, Kongens Lyngby, Denmark; [b]School of Design, Northumbria University, Newcastle upon Tyne, UK

ABSTRACT

Behavioural designers adopt practices from both design and behavioural science. Yet methodological discussions are fragmented across these fields. In response, this paper – for the first time – examines and draws together research on decision-making and methodologies applied to behavioural design via a systematic review. We identify three major themes challenging current understanding of behavioural design: *Complex behavioural design space*, *Systemic behavioural design*, and *Behavioural design empathy and ethics*. These themes give rise to a fundamental reconceptualization of the behavioural design process captured in our '*Behavioural Design IMPACT process model*'. This model integrates a *fuzzy front* and *back end* around a *co-evolutionary development process*. Through this, we explain how design and behavioural science practices can be synthesised to deal with *wicked behavioural problems* in a *complex context* where a long *intervention afterlife* is inevitable. This highlights the way towards more integrative behavioural design with major implications for researchers and practitioners across fields.

Trial registration

ClinicalTrials.gov identifier: NCT01602705.

ARTICLE HISTORY
Received 11 July 2023
Accepted 21 February 2024

KEYWORDS
Behavioural design; design methodology; design practice; decision making; systematic review

1. Introduction

The decisions designers take lay the foundations for both design processes and their outcomes (Andreasen et al. 1988; Blackhurst et al. 2004; Rodrigues et al. 2024). Effective decision making is thus essential to good design and has received substantial methodological support in the design literature, for example, in the form of the Delft Design Guide (Van Boeijen et al. 2020) and the Design Thinking Playbook (Lewrick et al. 2018). However, in the emerging field of behavioural design, designers combine decision making practices from both design and behavioural science with little clear direction on how to integrate these and access the best of both (Cash et al. 2017a; Voorheis et al. 2022).

CONTACT Camilla K. E. Bay Brix Nielsen ckeni@dtu.dk Technical University of Denmark, Department of Technology, Management and Economics, Produktionstorvet 424, 2800 Kongens Lyngby, Denmark

© 2024 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

Behavioural design tackles complex challenges through design and development of behaviour changing interventions (Bucher 2020; Niedderer et al. 2017), and is increasingly applied in areas including sustainability, health, and wellbeing (Barari et al. 2020, Fabio & Croce 2024; Schmidt & Stenger 2021; Tromp & Hekkert 2018). Behavioural design has a dual focus on: (i) achieving behavioural effect(s) (e.g. reducing speeding); facilitated by (ii) designing and implementing intervention(s) (e.g. information campaigns or speed bumps) (Khadilkar & Cash 2020). Hence, behavioural designers must not only consider and explore the design and development of various interventions but also realise the intended behavioural effect(s) facilitated by these.

This dual focus makes effective decision-making particularly challenging for behavioural designers, with far reaching implications for behavioural design projects and their outcomes. Specifically, there is a need for behavioural designers to balance design and behavioural science practices and mindsets to deliver successful interventions, with projects failing when either side is prioritised at the expense of the other (e.g. Aunger & Curtis 2016; Cash et al. 2022; Reid & Schmidt 2018; Voorheis et al. 2022). However, much of the current literature focuses primarily on behavioural theories and behavioural change techniques (BCTs), with design concerns narrowly related to translating these into 'delivery modes' at the level of e.g. 'paper based' versus 'digital' (French et al. 2014; Grudniewicz et al. 2015; Hargreaves et al. 2022; Michie et al. 2018). For example, Mertens et al.'s (2021) meta-analysis of over 200 nudging interventions concluded that effect size is significantly impacted by the selected BCTs, yet, a single BCT can be translated into endless contextually relevant design features that can critically impact success or failure. Consider how a 'paper based' (delivery mode) intervention 'translating information' (BCT) can be anything from a black and white letter in your mailbox to a colourful sign in a public space, and hence raise totally different design concerns (Nielsen et al. 2021). Here, several researchers have called for increased attention towards understanding under which conditions interventions succeed or fail (e.g. Bryan et al. 2021; Milkman et al. 2021; Szaszi et al. 2018; Szaszi et al. 2022). Thus, there is a critical need to better understand how design and behavioural science practices can be integrated to leverage the best of both to increase intervention success.

This need reflects calls for research from both the design (Cash et al. 2023; Nielsen et al. 2021; Schmidt 2022) and behavioural science literatures (Hagger & Weed 2019; Hansen 2018; Hughes & Ryan 2024; Patton et al. 2020). Therefore, we – for the first time – integrate understanding of decision making in behavioural design via a systematic literature review. Through this, we identify critical methodological interfaces between design and behavioural science practice and suggest means of integration. This has important implications for behavioural design as well as wider design and behavioural science research while pointing to several implications for behavioural design practitioners.

2. Theoretical background

Before we can address interfaces between design and behavioural science practices, it is first necessary to outline how these are currently understood in behavioural design and how methodological decision making can be understood more generally.

2.1. Decision making and methodology in behavioural design

A behavioural design project typically involves five overall process phases: (i) Identification and Framing of problem/solution space, (ii) Mapping and Description of contextual behavioural understanding, (iii) Framing and Development of Interventions, (iv) Iterative Testing and Refinement of intervention, and (v) Scaling up, Launching, and Maintenance (Cash et al. 2022). In each of these phases, and across the process as a whole, behavioural designers balance behavioural science and design concerns to deliver effective, explicable, and valuable solutions (Cash et al. 2022; Voorheis et al. 2022). Correspondingly, behavioural designers leverage methodological support drawn from both behavioural science and design (Bucher 2020). For example, abductive exploration of the problem-solution space – traditionally rooted in design – is critical to iteratively developing a rich and flexible understanding of the behavioural symptoms and their root causes (Tromp & Hekkert 2018; Van Boeijen et al. 2020), and deductive approaches – traditionally rooted in behavioural science – are essential to testing and validation of intervention effectiveness (Stegenga 2015). However, these methodological and decision making practices are currently primarily rooted in *either* design *or* behavioural science, with little support for their productive integration (Voorheis et al. 2022). Thus, despite the increasing acknowledgement that integration is essential for effective behavioural design (Nogueira & Schmidt 2022; Sanders et al. 2018), understanding of current integration challenges and possible means of mitigating these are extremely limited. To respond to this need, it is first necessary to understand the nature of design methodology and decision making practices in an operational way.

2.2. Understanding decision making and methodology

Designers enact professional disciplines through application of specific methodologies informing and supporting decision making (Daalhuizen et al. 2019; Günther & Ehrlenspiel 1999). In this way, methodology provides a theoretical backbone, which is then contextualised in practice with respect to factors including e.g. financial resources, time constraints, intuition, and prior experience (Kimbell and Street 2009; Kunrath et al. 2020). In this process, designers interact with an array of methodological elements ranging from design values and principles to concrete frameworks and methods (Lewrick et al. 2018, Van Boeijen et al. 2020). As such, to understand decision making and methodology it is necessary to examine all these elements and their interactions.

The recent work of Daalhuizen et al. (2019) provides a comprehensive overview of methodology and its multi-faceted relationship with design practice. Daalhuizen et al.'s (2019) Architecture of Design Doing (ADD) framework (Figure 1) distinguishes methodology with respect to two main levels of abstraction: 'Methodology for X' and 'Staging practice X'. Together, these structure eight main methodological elements (Table 1) relevant to decision making, offering an opportunity for the identification, analytical comparison, and integration of practices from diverse sources. As such, the ADD provides an appropriate lens for examining potential interfaces between design and behavioural science methodology and decision making as enacted in this multi-disciplinary behavioural design practice.

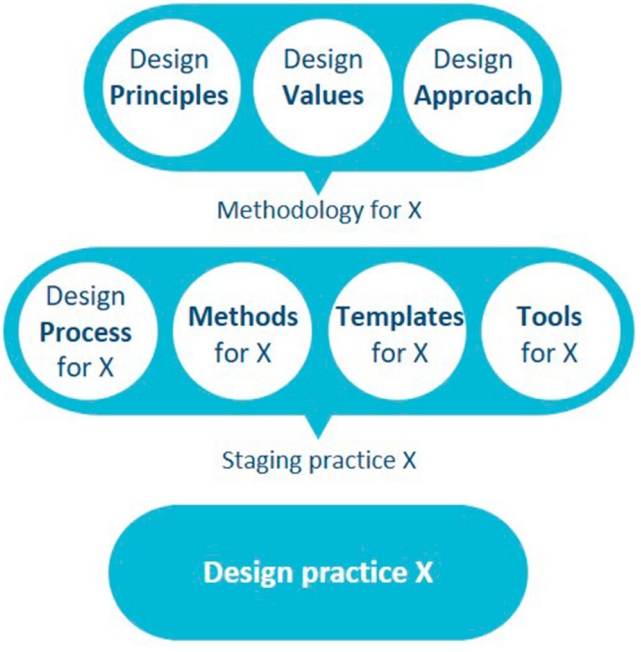


Figure 1. Architecture of Design Doing (ADD) (Daalhuizen et al. 2019).

Table 1. Definitions of elements in the ADD framework.

Element	Definitions quoted from Daalhuizen et al. (2019, 42–44)
Principles	'A design principle is a generic rule for design behavior that dictates a style of working for a certain practice or phase in a practice'
Values	'A design value is a generic statement for judgment in design that dictates or directs the decisions being made.'
Approaches	'A design approach describes a generic structure of design projects. As such, it describes basic activities that constitute any meaningful act of design.'
Processes	'A design process describes the abstracted phases and steps for a specific design practice intended to provide a comprehensive account of its structure.'
Methods	'A design method is a description of the structure of a specific activity, outlining the basic sequence or rules that make up single comprehensive activities within a practice.'
Templates	'A design template is a description of the structure of a design outcome, outlining the basic elements and lay-out for a deliverable.'
Tools	'A design tool is any piece of software or hardware that enables practitioners to perform their tasks or extends their capability to do so.'
Staging	'Staging refers to the use of information that links a methodological element to a specific context of use.'

3. Method

In answering the research need outlined in the introduction, we conducted a systematic literature review. Systematic reviews are used by multiple scholars' tackling similar research needs, e.g.: Stingl and Geraldi (2017) review on behavioural decision making in projects, Renzi et al.'s (2017) review on decision-making methods in engineering design, and Hay et al.'s (2017) review on conceptual design cognition. Following Hay et al. (2017) the PRISMA 2020 statement (Page et al. 2021) informs our literature review. The systematic review

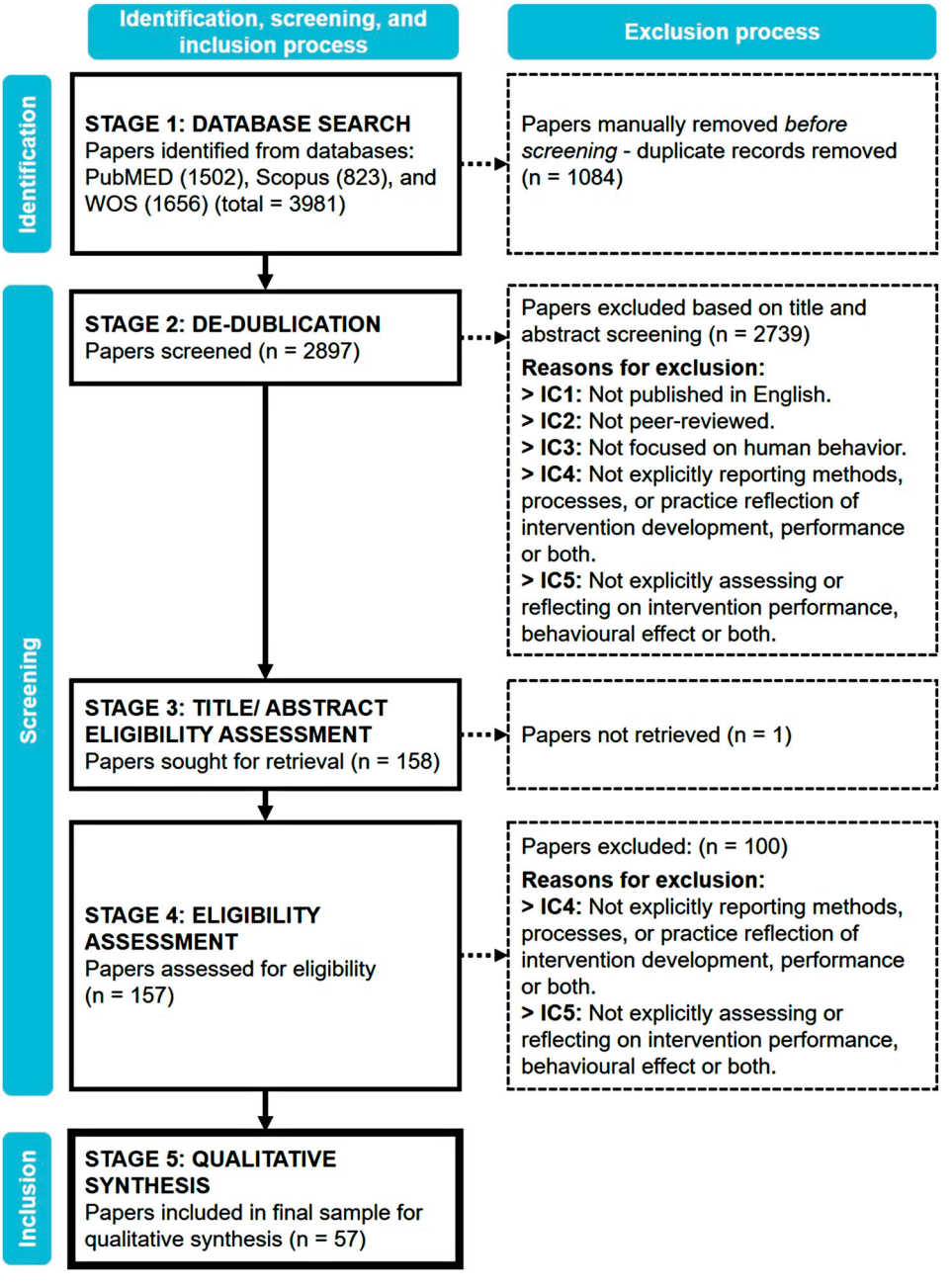


Figure 2. PRISMA flow diagram (type: for new systematic reviews including database searches).

process is summarised in Figure 2 and included three main phases: *identification (Stage 1)*, *screening (Stage 2-4)*, and *inclusion (Stage 5)* of relevant scientific papers in our final sample. Three research assistants (R1, R2, and R3) and three researchers (R4, R5, and R6 – the authors of this paper) co-worked to complete this process. R1-R3 are educated in design, including behavioural design, and R4-R6 are design researchers with expertise in design, behavioural design, design practices, and design thinking.

3.1. Identification, screening, and inclusion of scientific papers

R1-R3 collected papers between March 3rd 2022 and April 19th 2022. The search string is structured with Behaviour/behavior∗ as the overall domain, together with six AND themes, of which three contained OR alternatives (Interven∗, Change∗, Design∗, Effect/Quality/Outcome/Metric/Value, Anticipat∗/Determine/Measure/Assess, and Human/People), see Table 2. Together, the search string reflects various aspects of decision making in behavioural design. This search string was developed in agreement across all six research team members.

The search string was applied at title and abstract level across three major scientific databases: PubMED, Scopus, and Web of Science (WOS), where most behavioural design work is published. The search was limited to English peer-reviewed papers published in journals with publication years between 2014-2022. From the database search (Stage 1), R1-R3 obtained 3981 papers of which 1084 were removed in the deduplication process (Stage 2). The remaining 2897 papers were screened at the title and abstract level (Stage 3) in three rounds. Round (1) R1-R3 each screened 1/3 of the 2897 papers using the inclusion criteria (Table 3). Here, the papers were assigned to either of three categories: 'Exclude now' (papers that did not meet a single inclusion criteria), 'Include at this stage' (papers that clearly met all inclusion criteria), 'Screen again' (papers that were less straight forward to put in either of the remaining two categories). Round (2) R1-R3 discussed their preliminary findings between each other. Round (3) R1-R3 discussed and finalised their findings with R4 and R5 (the first and second author), such that all papers at this point were assigned either 'Exclude now' or 'Include at this stage'. After these three rounds, 158 papers remained and were sought for retrieval, of which one could not be retrieved. The remaining 157 papers were then assessed for eligibility at full text level (Stage 3) again in three rounds, following the same structure as in the preliminary screening: R1-R3 each screened 1/3 of the papers (using the inclusion criteria in Table 3), then first discussed their findings between each other, and next discussed again and finalised with R4 and R5. After this, 57 papers remained

Table 2. Search string for systematic literature review.

Domain	Themes ← AND →					
Behaviour*/ behavior*	Interven*	Change*	Design*	Effect **OR** Quality **OR** Outcome **OR** Metric **OR** Value	Anticipat∗ **OR** Determine **OR** Measure **OR** Assess	Human **OR** People

Table 3. Inclusion criteria.

No.	Inclusion criteria (IC)
IC1	Paper must be published in English
IC2	Paper must be peer-reviewed journal article
IC3	Paper must focus on interventions targeting human behaviour (excluding animal behaviour)
IC4	Paper must explicitly mention either methods, processes, or reflective practices related to intervention development, intervention performance, or both
IC5	Paper must explicitly assess or reflect on intervention performance, behavioural effect, or both

and were included in the final sample as the basis for our analysis (Stage 5). Figure 2 reports the details of the systematic review process.

3.2. Sample characteristics

In total, 57 papers were included in the final sample (Table A1 in appendix). These papers date from 2014 to 2022, with 28% published in 2014-2016, 30% published in 2017-2019, and 42% published in 2020–2022 (April 19th). In order to provide an overview of the type of work conducted we categorised the 57 papers into four categories according to the following descriptions:

i) ***Intervention study papers*** describing design, development, implementation, and/or testing of specific interventions (14 = 25%)
ii) ***Review papers*** describing either systematic, rapid, or scoping reviews with the primary purpose of presenting theoretical frameworks for specific interventions or specific intervention domains (19 = 33%)
iii) ***Protocol method papers*** describing a proposed protocol for intended work processes of either scoping reviews, or design, development, implementation, and/or testing of specific interventions (8 = 14%)
iv) ***Theory papers*** describing original knowledge frameworks, methods, and work approaches for conducting behavioural design (16 = 28%)

3.3. Qualitative synthesis approach

Due to our focus on behavioural design decision making and the lack of current theory in this area, we do not attempt quantitative meta-analysis and instead adopt a qualitative synthesis approach aimed at eliciting understanding of the decisions and practices enacted during behavioural design. This approach is similar to Hay et al.'s (2017) systematic review of design cognition. Our qualitative synthesis was based on a thematic content analysis in multiple rounds. Adapting an initial explorative approach, in parallel R1-R3 and R4 compiled a list of keywords broadly connected to decision making across the 57 papers in the final sample in two separate documents. Forming the basis for content analysis, R1-R4 presented the two lists to R5 and R6. R4-R6 then worked iteratively to synthesise and structure findings in themes. Throughout, Daalhuizen et al.'s ADD framework was used as the analytical basis (Figure 1) and guided evaluation of methodological elements.

4. Findings

The findings are structured according to the six methodological elements in Figure 1 and summarised in Table 4 at the end of this section.

4.1. Design Principles: Imbalanced combination of deductive and abductive principles

The main insight regarding design principles was a dominant use and detailed reporting of deductive working styles in combination with frequent, yet superficial reporting of abductive working styles. Specifically, deductive principles were dominant in the definition of

behavioural problem space and behavioural goals while abductive principles were connected to descriptions of interventions, embodiment of design features, and decision making connected to intervention development.

In terms of deductive principles, the majority of the reviewed papers treated both the behavioural problem space and the behavioural goal as well defined. Here, many papers introduce the behavioural problem as pre-defined and constrained by existing theory or knowledge. For example, the design aim reported in Preece et al. (2021, 3) implies treating the design problem as well defined and constrained: '… *create a behaviour change intervention for knee OA (osteoarthritis) which was suitable for delivery by an appropriately trained physiotherapist within a UK NHS outpatient clinic*'. Further, many papers introduced the design goal in the early development work and presented them as a pre-defined outcomes in the form of overall intended behavioural effect (e.g. 'reducing intake of sugar sweetened beverages' (Vargas-Garcia et al. 2015)) or 'maintaining physical activity of post stroke patients' (Masrur et al. 2020). These pre-defined outcomes are rarely iterated or unpacked, except for a small number of papers explicitly mentioning a design activity of 'defining the outcome in behavioural terms' (Step 1 in Michie et al.'s (2014) 8 step intervention development process) e.g. defining 'gain bodymass by 5.6 kg' in behavioural terms: 'consumption of 25.1 MJ high quality food stuffs each day for 12 weeks' (Costello et al. 2018).

This focus and depth of reporting of deductive working principles was, however, contrasted by frequent but incomplete reporting of abductive principles. For example, interventions across the reviewed *intervention studies* were described in terms of Michie et al.'s (2013) 93 BCTs, ranging from combinations of three (e.g. '*shaping knowledge & natural consequences*', '*goal setting & action planning*', '*self-monitoring*' (Liew et al. 2021)) to 11 BCTs (Patton et al. 2020). Here, despite BCTs being present in all *intervention studies,* the ideation associated with embodying the BCTs as actual design features was not documented, suggesting that such ideation lacked documentation or might be neglected all together. Further, reporting of design exploration and abduction was typically fragmented and often characterised as outcomes from deductive activities. For example, Liew et al. (2021, 85) describe choosing three BCTs '*based on clinical and patient consultations, and relevant research evidence*' without reporting further information. A small number of papers elaborate further by reporting chosen 'delivery modes' (e.g. Beall et al. 2014) report that their intervention shall be delivered '*face-to-face*' and '*print-based*' hinting at overall embodied design features. However, additional details of the decision making process of chosen design features, including delivery modes, were often not explicitly reported but, at best, implied – e.g. '*given high rates of internet usage in UK and similar development countries, it was felt a digital platform would maximise reach and accessibility*' (Knight et al. 2021, 105). As such, despite the centrality of this contextual design work it is difficult to dissect what research evidence and how various stakeholders informed the decision making process, as well as why specific BCTs, delivery modes, or design features were chosen or developed over others. Thus, explicit and detailed reporting of design concerns and decision making was often lacking, in spite of being critically relevant for later stage iteration and reflection on intervention effectiveness.

Bringing these findings together, there is a clear imbalance in the dominant reporting of deductive principles at the expense of abductive principles. Overall, behavioural problems and goals are introduced early in the development phase, treated as well-defined, little to not explored, and are rarely – if at all – re-visited or revised. Therefore, there is a high-risk

of missing potentially valuable solution directions that might be explored if design exploration and abduction would have happened. In addition, reporting of intervention design is typically limited to sets of BCTs with overall descriptions of delivery modes only hinting at the design work done regarding actual embodiment of design features. These discrepancies in the work done and the work reported highlight the need to better understand how behavioural designers can apply and report a more comprehensive set of integrative principles.

4.2. Design values: Prioritisation of theory- over stakeholder-informed decision making

The main insight regarding design values was that theory-informed decision making was documented in detail. At the same time, stakeholder-informed decision making was little reported, despite playing a major role across the review sample. Specifically, theory-informed decision making took priority in constraining the design space and identifying intervention features and was typically reported in depth. In contrast, various stakeholder-informed decision making was frequently used in connection to decision verification, intervention refinement, and performance assessment, yet, typically superficially reported.

All of the papers in our sample report theory-informed decision making as central to the process (e.g. constraining the problem-solution space by compiling existing knowledge through literature reviews (Atkins et al. 2020; Goyal et al. 2016; Jovicic et al. 2015; Perski et al. 2022). In connection to this, rapid, scoping, or systematic reviews were a key part of decision making during intervention development in almost all papers. Further, multiple papers strove for replication of interventions, either by i) explicit documentation for future replicability (Band et al. 2017), or (ii) identification of whole interventions or intervention elements suitable for replication identified through literature reviews. These findings simultaneously highlight the prioritisation of theoretical and scientific rigour and minimisation of contextual factors and opportunities of innovation, suggesting a tension between design values versus behavioural science values.

This insight was further supported by the way stakeholder-informed decision making was reported. Here, many of the reviewed papers reported involving a variety of stakeholders (e.g. including intervention facilitators, intervention users, or knowledge experts) to help navigate in the problem-solution space after it already had been constrained by theory (as outlined in the previous paragraph) (e.g. Knight et al. 2021; Preece et al. 2021). Further, Goyal et al. (2016, 97) report user-involvement only as an illustration in their development process model; Barnett et al. (2014, 5) report in one sentence '*Four experts judged which techniques [BCTs] would be effective*'; and Knight et al. (2021) imply the use of stakeholder-informed decision making in a late-stage intervention refinement loop by stating that participant feedback led to a final iteration of the intervention. However, none of these works position stakeholder engagement in a leading or balanced role, nor report its impact in depth compared to the reporting of theory-informed decision making. Often authors did not commensurate to its role in the project. Overall, there was a general trend in using stakeholder engagement to confirm or validate theory-informed decisions rather than as basis for more 'designerly' stakeholder-informed decision making, that is, explorative, abductive, and open-ended stakeholder interaction (Cross 2006).

In sum, while both theory and stakeholder engagement are found in the majority of papers in our sample they are neither treated or reported equally. This highlights a prioritisation of theory-informed decision making and its associated values, and positions stakeholder engagement in a confirmatory, secondary role which is at odds with its extensive usage in the design and innovation literature (Lewrick et al. 2018; Van Boeijen et al. 2020). However, despite this value difference, many projects relied on stakeholder involvement to shape key aspects of the project. Thus, this finding emphasises a critical disconnect between the expressed values and documented practice.

4.3. Design approaches: outcome- over application-focused intervention assessment

The main insight with respect to the design approach was a dominant use of outcome-focused over combined outcome- or application-focused intervention assessment approaches. Most papers in our sample reported intervention assessment primarily in terms of behavioural outcomes and only sporadically reported application assessment in terms of how the intervention was 'received' by users.

In terms of outcome-focused assessment, the majority of papers in our sample assessed intervention performance by linking selected BCTs with various effectiveness measures focused on the behavioural goal. For example, Schroé et al. (2020) assessed efficacy – measured by time spent on a physical activity – to compare four interventions embedding either none, one, two, or three BCTs. Further, in a systematic review and meta-analysis, Kassavou & Sutton (2018) assessed intervention effectiveness – measured by level of medical adherence – across multiple automated telecommunication interventions based on three intervention features (i) format (e.g. 'text' or 'voice'), (ii) duration, frequency, and direction of messages, and (iii) embedded BCTs. In assessing intervention performance, especially in more recent papers, increased attention is given to addressing potential overlap of intervention features across comparison groups, e.g. embedded BCTs (Meade et al. 2019) or delivery formats (Kassavou & Sutton 2018), but rarely to the contextual design of the intervention embodiment itself. As such, critical information is often lacking regarding design features that might significantly impact performance in social and physical contexts (e.g. user experience). Despite this outcome focus, papers reported using a variety of quantitative (e.g. Reynolds et al. 2019; Kassavou & Sutton), qualitative (e.g. Bowers et al. 2020), and mixed-methods approaches (e.g. Schroé et al. 2020). For example, Bowers et al. (2020) report conducting in depth qualitative synthesis of 'think aloud' interviews. This hints at a disconnect between a mono-dimensional focus on outcome assessment and a more varied utilisation of both design and behavioural science approaches.

This finding was further emphasised by the limited examples of application-focused assessment. Here, several recent papers complement their outcome-focused assessment with an emphasis on 'fidelity' (e.g. Liew et al. 2021; Patton et al. 2020). Fidelity here refers to delivering and facilitating the intervention in an intended and consistent way across (i) multiple interventions and settings (e.g. across clinics), (ii) multiple intervention facilitators (e.g. clinical professionals), (iii) or both. Further, a number of recent papers extend this consideration of fidelity to include user acceptability and adherence (e.g. Palsola et al. 2020; Preece et al. 2021), as interventions delivered with high fidelity can still underperform due to low

user acceptability or adherence (Palsola et al. 2020). These works highlight how intervention performance is often determined by specific contextual, embodied design issues rather than general underlying BCTs, and hence the need to expand and integrate assessment approaches.

To sum up, there is a tradition in behavioural science of outcome-focused assessment using a variety of data collection and analysis approaches. However, this is challenged by an increasing awareness of design related nuances in intervention performance that demand the combination of outcome- and application-focused assessment. This emphasises a need to better understand how behavioural designers can holistically combine multiple types of assessment mindsets and approaches.

4.4. Design processes: dominant linearity

The main insight regarding design processes was a dominant use of linear development processes with sporadic 'within phase' iteration. Specifically, most papers reported a linear development process emphasising deductive work where the output of one process phase is the primary input for the next process phase.

The majority of both *intervention studies* and *protocol methods* in our sample employed Michie et al.'s (2014) 8 step intervention development process (e.g. Costello et al. 2018; Patton et al. 2020), the Medical Research Council's (MRC) framework (Shahsavari et al. 2020) (e.g. Goyal et al. 2016; Liew et al. 2021), or other linear development process (e.g. Knight et al. 2021; Pincus et al. 2015 ; Preece et al. 2021). Common to all these applications was the consistent use of linear phases without iterative loops. In addition, two papers applied a Theory-, Evidence-, and Person-based Approach (PBA) where the process reflects two, parallel, linear work streams (Band et al. 2017; Bowers et al. 2020). However, this still primarily emphasises one-way deductive development work.

Despite the overall linearity in reported development four *intervention study* papers explicitly reported minor iteration within a design phase (e.g. Liew et al. 2021). Examples include (i) iterative refinement of intervention design (Bowers et al. 2020), (ii) iteratively designed feedback (Barnett et al. 2014), (iii) iteratively developed components (Preece et al. 2021), and (iv) iterative intervention development and evaluation (Band et al. 2017). Notably, only Band et al. (2017, 11) explicitly reported elements of iteration across process phases:

> … if there is insufficient time to complete intervention planning before intervention development must commence (as in the case of HOME BP), then preliminary theoretical modelling can be based on partial evidence, and then updated and refined once evidence collation and analysis is complete.

In these examples, iteration was needed to mitigate the risks of bringing faulty assumptions or poor design decisions forward from one phase to another and meet the demands of flexibility in developing interventions fit for complex contexts. However, as description of iteration was sparse, it was difficult to assess how iteration decisions were made and managed in practice.

Together, these findings emphasise the widespread use of linear processes and general lack of iteration. This is at odds with extensive usage of iteration in the design and innovation literature (Adams & Atman 1999; Roozenburg 1993). As such, there is a disconnect

between current processes and the actual, more iterative demands of behavioural design. This reveals a need to better understand how behavioural designers can integrate iterative, (re)framing and co-evolution approaches – within and across – process phases in complex intervention development.

4.5. Design methods & templates: a dominant scientific approach

The main insight regarding design methods and templates was a dominant 'scientific' approach to their use, here referring to deductive approaches emphasising internal validity. This was in stark contrast to the sporadic use of designerly approaches emphasising contextual adaptation and external validity.

Here, the majority of the papers in our sample reported using well-established methods and templates in a manner equivalent to more traditional scientific work, implying deductive use, with a focus on documenting the achieved outcome. Examples include: (i) scoping, rapid, or systematic reviews supporting front-end decision making (e.g. Goyal et al. 2016; Preece et al. 2021); (ii) templates such as TiDiEr (Hoffmann et al. 2014) for intervention documentation (e.g. Liew et al. 2021; Schroé et al. 2020); and (iii) checklists such as APEASE (Michie et al. 2014) for intervention design description and decision making (e.g. Patton et al. 2020; Perski et al. 2022). These were all primarily framed as increasing internal validity, and comparability across interventions, however, they only capture a fraction of the actual design work undertaken in each case.

Here, only a few papers reported more designerly use of methods and templates in support of the major creative and development work conducted, however, details of such use were sparsely – if at all – described. For example, Preece et al. (2021, 6) report a design activity called '*exploration of patient beliefs*' where they '*encourage patients to contextualise their own pain experiences*'. Hence, Preece et al. (2021) imply applying an abductive method involving patients, which – even though not explicitly reported – presumably influenced their decision making. As an exception in our sample, Patton et al. (2020), explicitly focus on reporting design decisions in relation to outcomes and contextual factors influencing choices throughout intervention development, arguing that increasing decision making transparency will make exploration and explanation of intervention (in-)effectiveness easier. This illustrates the imbalance inherent to current method use, where some methods are perceived as scientific and reported in detail while others are underreported despite their key roles in supporting intervention development.

In sum, these findings highlight a focus on traditional scientific style reporting of method and template use and a seeming reticence to apply the same level of reporting to equally important but more flexible methods application from design. This, points to an urgent need to develop a more consistent approach to method documentation commensurate with the impact of both designerly and scientific method use.

4.6. Design staging: detailed staging of 'scientific' work versus simple staging of 'design' work

The main insight regarding design staging was the highly detailed reporting of staging of conventional scientific work, in contrast to superficial reporting of staging associated with

more designerly focused work. This follows from and compounds the disparity in focus on method reporting highlighted in the prior section.

Exploring this insight further, most papers in our sample reported the staging of (i) the scientific study, (ii) outcome-focused intervention development, (iii) or both, in detail and at length. Specifically, almost all papers in our sample described the methodological choices related to the scientific study in dedicated method sections with the *review* and *protocol method* papers being almost entirely focused on scientific methodology. For example, review related activities were described in detail using to the PRISMA guidelines in dedicated sections (e.g. Knight et al. 2021; Perski et al. 2022), except in one *intervention study* (Preece et al. 2021) where the review formed one out of many intervention development activities. Thus, a general emphasis was given to reporting outcomes and measures chosen to assess intervention effectivities (e.g. Vargas-Garcia et al. 2015), or the results of pilot testing (e.g. Westland et al. 2017).

In contrast, reporting of the design staging and associated decision making was typically neglected. Specifically, there was a general lack of both description of process related design decisions and considerations, as well as detailed description of design features embodied in interventions. When descriptions were present, they were often fragmented and provided in just one or two sentences. For example, Band et al. (2017, 9) state that '*The HOME BP logic model underwent several iterations incorporating team members and other stakeholder feedback*', however this description lacks contextual information on who, when, where, how, and why. Similar information was also lacking in Knight et al.'s (2021, 105) description of '*the research team worked with a digital development company to design and build a conversational interface*' and later '*once developed, the digital intervention was piloted with 18 members of the public (nine male/nine female) who had no previous experience*'. A similar pattern was observed in staging design choices related to the intervention development process. For example, Barnett et al. (2014) design and compare three interventions and describe the included intervention components as '*educational materials*', '*feedback on practice rates of high-risk prescribing*', and '*a theory-informed intervention embedded in the feedback*'. Here, Barnett et al. (2014, 3) provide staging information on who participated '*All educational materials were created by the research team in collaboration with the Advisory Group and where necessary topic-specific experts*', but offer no further description of intervention design. Only Patton et al. (2020) dedicate substantial space to documenting ongoing elaboration of alternatives and considerations leading to their final design choices. This highlights how the reporting of more traditionally scientific activities is prioritised over – often very substantial yet underreported – design work.

Together these findings illustrate a discrepancy of how traditional scientific work practices versus traditional design work practices are considered and reported, despite both being essential in all projects. This directly points to the need to better understand what aspects of staging that should be consistently reported across the whole behavioural design project.

4.7. Summary of findings

The findings are summarised in Table 4, which provides examples of an overall, critical imbalance between the prioritisation and reporting of traditionally scientific- and design-related work and its associated decision making.

Table 4. Summarised findings from literature review including examples from sample.

ADD element* & main insight	Overall findings from review	Examples of findings in review sample
Design principles Imbalanced **deductive** and **abductive** working principles for problem-solving	**Deductive problem solving:** Treating behavioural problem spaces as well-defined Treating behavioural goals as well-defined	Pre-defining behavioural problems, behavioural goals, or both: • E.g.: '… create a behaviour change intervention for knee OA (osteoarthritis) which was suitable for delivery by an appropriately trained physiotherapist within a UK NHS outpatient clinic' (Preece et al. 2021) • Describing behavioural goals as overall behavioural effect: • E.g.: '… promoting short-, moderate-, and long-term physical activity maintenance among patients post-stroke' (Masrur et al. 2020)
	Abductive problem solving: Describing embodied design features simplistically	Describing or treating interventions as sets of BCTs (E.g.: Bondaronek et al. 2019; Liew et al. 2021; Patton et al. 2020; Schroé et al. 2020) Describing design features in keywords: • E.g. Barnett et al. (2014, 3) 'educational materials'; 'feedback on practice rates of high-risk prescribing'; and 'a theory-informed intervention embedded in the feedback'.
Design values Prioritising **theory-informed** decision making over **stakeholder-informed** decision making	**Theory informed decision making:** Constraining intervention development space based on theory (often through literature reviews)	Constraining problem space by behavioural theory reviews (e.g.: Goyal et al. 2016; Jovicic et al. 2015; Masrur et al. 2020) Constraining solution space by replicable intervention features review (e.g.: Band et al. 2017)
	Stakeholder informed decision making: Constraining intervention development based on stakeholder's knowledge and input	Informing decisions based on input from intervention facilitators, intervention users, or knowledge experts: • E.g.: 'Four experts judged which techniques [BCTs] would be effective' … (e.g.: Barnett et al. 2014, 5)
Design approaches Outcome over **application**-focused decision making approaches	**Intervention outcome-focused decision making:** Testing and implementation of intervention focuses on behavioural outcome	Assessing intervention performance by level of achievement of behavioural goal (e.g.: Harries & Rettie, 2016; Johnson & May, 2015; Schroé et al. 2020; Westland et al. 2017)
	Intervention application-focused decision making: Testing and implementation of intervention focuses on application	Assessing intervention performance by intervention application including: • Fidelity (e.g. Patton et al. 2020; Westland et al. 2017) • Acceptability (e.g. Preece et al. 2021; Westland et al. 2017) • Adherence (e.g. Palsola et al. 2020; Westland et al. 2017)
Design processes Dominant use of **linear** design processes combined with few **iterative** activities	**Linear design process models:** Structuring design and development work linearly or parallel linearly with forward directed design phases	Linear process models applied: • Michie et al's (2014) 8 step intervention development process model (e.g. Costello et al. 2018; Patton et al. 2020) • Medical Research Counsil's (MRC) framework (e.g. Goyal et al. 2016; Liew et al. 2021) • (Parallel linear structured) The Theory-, Evidence-, and Person-based Approach (PBA) (Band et al. 2017; Bowers et al. 2020)
	Iterative process activities: Reporting iterative within phase activities	Iterative intervention functionality optimisation (e.g. Barnett et al. 2014)

(*continued*).

Table 4. Continued.

ADD element* & main insight	Overall findings from review	Examples of findings in review sample
Design methods & templates Dominant '**scientific**' application and sporadic '**designerly**' application	**'Scientific' application:** Applying methods and templates deductively and 'more rigidly' (often focusing on internal validity, reliability, and comparability)	• Literature reviews (e.g. Band et al. 2017; Goyal et al. 2016; Knight et al. 2021; Preece et al. 2021) • TiDieR template (e.g. Meade et al. 2019; Schroé et al. 2020) • APEASE checklist (e.g. Patton et al. 2020; Perski et al. 2022)
	'Designerly' application: Applying methods abductively and 'more flexibly'	Adaption of standardised methods (e.g. Preece et al. 2021) Using 'unspecified' methods (e.g. Patton et al. 2020)
Design staging Detailed staging of '**scientific**' work versus simple staging of '**design**' work	**Detailed staging of 'scientific' work:** Describing methodological choices in scientific study, behavioural outcomes of intervention, or both in rich detail	Dedicated method sections focusing on 'scientific' work (e.g. Knight et al. 2021; Masrur et al. 2020, Perski et al. 2022) Detailed description of behavioural intervention effectivities (e.g. Schroé et al. 2020; Vargas-Garcia et al. 2015) Detailed description of intervention pilot testing results (e.g.: Preece et al. 2021; Westland et al. 2017)
	Simple staging of 'design' work: Describing process related design work fragmented	Describing design work in fragments of 1–2 sentences, e.g.: • "The HOME BP logic model underwent several iterations incorporating team members and other stakeholder feedback' (Band et al. 2017) • 'All educational materials were created by the research team in collaboration with the Advisory Group and where necessary topic-specific experts' (Barnett et al. 2014, 3)

[a] Adaption of Daalhuizen et al.'s (2019) Architecture of Design Doing (ADD) applying design principles, values, approaches, processes, and staging as originally clustered in ADD, putting 'design methods' and 'design templates' under one methodological element, and excluding 'design tools'.

5. Discussion and implications

Our findings hold several important implications for behavioural design research and practice. Specifically, we found that while design and behavioural science working styles and associated decision making practices were sometimes at odds, they are more often highly complementary yet unequally treated and reported, as summarised in Table 4. This substantially extends and elaborates calls to better understand this interaction from across literatures (Cash et al. 2022; Hagger & Weed 2019; Patton et al. 2020; Schmidt 2022), and provides a foundation for drawing out and reporting the best of both design and behavioural science approaches (Reid & Schmidt 2018; Voorheis et al. 2022). Hence, in this section, we work towards an integrated understanding based around three main themes: *Complex behavioural design space*, *Systemic behavioural design*, and *Behavioural design empathy and ethics*.

5.1. Complex behavioural design space

The first theme highlights the complexity of the behavioural problem-solution-space (primarily linked to findings in sections 4.1, 4.2, 4.4, and 4.6).

5.1.1. 'Wicked' behavioural problems and the inevitable fuzzy front end

Every intervention project in our sample dealt with some form of 'wicked problem'; where behaviour was linked to multiple, multifaceted causes intertwined in a complex and

dynamic system (e.g. Bell et al. 2020; Smith et al. 2016). Hence, a productive understanding of the behavioural problem will inevitably depend on both the perspectives applied during problem exploration, and co-evolve with increased knowledge during development (Cash et al. 2020; Dorst 2006; Dorst & Cross 2001). The early stage of exploring such 'wicked problems' is often called the fuzzy front end (Reinertsen 1985; Smith & Reinertsen 1998), acknowledging that the problem cannot be completely defined up-front. This is at odds with most behavioural design practices reported in our sample, which typically described starting points of well-defined behavioural goals (e.g. Pincus et al. 2015; Preece et al. 2021) implying that a relevant, solvable, and relatively stable problem could be clearly identified and defined up-front. For example, Michie et al.'s (2014) widely applied eight step process model presents defining and translating the problem into behavioural terms as a linear, deductive prerequisite before behavioural diagnosis. Further, in most projects there was no reporting of how – or even if – reframing was applied. This misalignment between the wicked nature of behavioural problems and predominant behavioural design practice means that without addressing the fuzzy front end, behavioural designers risk not only solving 'wrong problems' but also attempting to solve them in 'wrong ways'.

To resolve this misalignment, parallels can be drawn with design thinking and radical innovation practices specifically tailored to deal with wicked problems (Buchanan 1992; Waddock et al. 2015). For example, the integration of abductive exploration and problem reframing with more deductive development is an essential tenet of design thinking and embedded in many design and innovation processes emerging from a fuzzy front end (Dunne & Martin 2006). This points to an overall need to integrate abductive exploratory practices to productively move through the fuzzy front end, identifying and situating more specific behavioural problems that can then be considered, defined, and resolved. Here, valuable steps in Michie et al.'s (2014) 'behavioural diagnosis' would benefit from being combined with abductive exploration methods traditionally rooted in design, such as 5xWhy (Ohno 1988), chunking (Sy 2007), behavioural storyboards (Martin & Hanington 2012), and even brainstorming (Osborn 1953) on behavioural assumptions. This would allow behavioural designers to identify multiple relevant behavioural root causes and reduce the risk of targeting behavioural symptoms, prior to more time-consuming systematic activities (such as (dis-)proving of behavioural assumptions and increased behavioural knowledge through theoretical exploration and user research). Hence, by acknowledging the wickedness of behavioural design problems and the associated need to deal with a fuzzy front end of behavioural design, research is presented with a conceptual model for the overall integration of design and behavioural science practices. This provides the basis for three key implications for practice:

1) Behavioural design problems are **wicked**, dealing with multiple, multifaceted, and intertwined potential problem causes. Behavioural designers should avoid considering behavioural design problems as 'tame problems'; assuming that behavioural problems can be easily defined and solved in well-defined contexts leads to a high-risk of solving 'wrong problems' – e.g. symptoms rather than root causes –, solving them in 'wrong ways', or both.
2) Wicked problems demand **fuzzy front end** approaches combining abductive exploration and reframing with more deductive development work. Behavioural designers should employ abductive problem exploration prior to more resource intensive

deductive contextual research to decrease risk of solving 'wrong problems', solving them in 'wrong ways', or both.

3) Fuzzy front end approaches highlight the importance of **documenting iterative reframing activities**. Decision making linked to small and large scale (re)framing can have a significant impact on the project process and outcomes. Hence, behavioural designers should document decision making essential to understanding both co-evolution of problem and solution as well as intervention design and performance.

5.1.2. Complex behavioural solutions

Most projects in our sample employed combinations of interventions working together in a system to support behaviour change – e.g. a health care package including both consultations with doctors, a patient app, information materials, text message prompts etc. (e.g. Masrur et al. 2020; Patton et al. 2020). These interventions – both separately and on a systems level – imply a multitude of crucial interaction points both affected by and affecting different users' behaviour patterns (consecutive decisions, (Miltenberger 2016)) formed by heuristics and biases (Lockton 2012; Tversky & Kahneman 1974). Thus, intervention design implies an almost unlimited scope for combination of embodied design features and BCTs. However, we see no explicit reporting of abductive exploration or creativity techniques connected to intervention design. On the contrary, reported approaches typical described either (i) selecting BCTs (e.g. Liew et al. 2021) or (ii) simply stating that something had been developed (e.g. Knight et al. 2021) both with no ideation reported on translating BCTs into embodiment design features. As such, intervention design was typically treated – or at least documented – as a more deductive selection activity associated with BCTs isolated from the major complexity of the intervention and its implementation context. This misalignment in scope and approach essentially neglects design embodiment and precludes effective utilisation of the full intervention design space.

Here, abductive approaches and creativity techniques logically complement deductive work by facilitating a more thorough, reflective, and agile exploration of the available design space. For example, rapid prototyping, stakeholder feedback, co-design, and design experiments can all provide directional insights (Bate & Robert 2007; Dunne & Martin 2006). These can help ensure effective targeting of later (and more substantial) trials or other testing programmes, and account for both design and behavioural considerations. Such activities are not only valuable during intervention development but across the entire behavioural design process, by updating and aligning decisions through iteration and refinement (Lawson 2006). These activities also help behavioural designers to better evaluate how specific design choices or changes will affect the entire intervention system prior to full scale testing or implementation (Maier & Cash 2022). In conjunction with the insights from Section 6.1.1 this dictates a systematic yet flexible approach that blends aspects of design and behavioural science approaches. This has two main implications for practice:

4) Interventions reflect **complex combinations** of embodied design features and BCTs. Behavioural designers should deploy abductive exploration and creativity techniques in collaboration with relevant stakeholders to help scope, and iteratively refine understanding of these possible combinations.

5) Interventions should – almost always – be understood as **complex and comprehensive**, comprising multiple, intertwined elements acting as an internal intervention system. Behavioural designers should acknowledge that design features of each intervention element will affect the entire internal intervention system and thereby affect the collective effectiveness in changing behaviour. To mitigate this, exploratory prototyping and similar activities should be used to unpack possible interactions prior to committing to a final intervention approach.

5.2. Systemic behavioural design

The second theme highlights the systemic, longitudinal nature of behavioural design (primarily linked to findings in sections 4.3, 4.4, and 4.5).

5.2.1. Contextual interventions and the inevitable fuzzy back end

Most projects in our sample employed combinations of interventions working together to support behaviour change. This shifts the conceptualisation of behavioural design from a single project to an evolving system framing, where it is necessary to consider how individual interventions and the collective intervention system (mis-)align internally and externally with existing infrastructure (Alexander 1971; Koohsari et al. 2015; Losinski et al. 2014). For example, what might seem a small detail in embodied design, such as an interventions' font size (Chang & Brainerd 2022) colour (Jalil et al. 2012), or sound (Meng & Kang 2016), can impact compliance with existing infrastructure, such as physical placement or timing of intervention. This way of thinking is gaining increased attention across multiple design fields including engineering design (Jiao et al. 2022), systems design (Maier & Cash 2022), and UX (Jiao et al. 2017; Wang & Jiao 2022). This shift in conceptualisation highlights the need to consider when to challenge existing infrastructures or status quo and increase innovation (Boland & Collopy 2004), as well as the need to consider user acceptance and broader interaction with an intervention in context over time (Lin & Mai 2018; Müller-Riemenschneider et al. 2008; Rosenberger et al. 2017). However, these considerations were not reflected in most development processes and measures of intervention performance applied in our sample, in contrast, it was typically assumed that a relevant, solvable, and relatively stable implementation context could be clearly identified and defined up front. In addition, measurement is primarily concerned with intervention effectiveness (e.g. Costello et al. 2018). Specifically, understanding contextual factors' impact on intervention performance depends on the perspectives taken during implementation exploration, and will co-evolve with increased knowledge during development, implementation, and follow-up assessment as known from e.g. Life Cycle Analysis (LCA) (Petrillo et al. 2016; Stavropoulos et al. 2016). This creates a fuzzy back end, where – as with the fuzzy front end – contextual factors of implementation in existing infrastructures cannot be completely defined up-front.

In recent years, increased attention has been paid to the need for explorative and creative approaches in the fuzzy back end (Almqvist 2017), with iterative assessment and refinement of intervention(s)' performance – not only through 'desk' design activities but also 'in the field' (Aunger & Curtis 2016; Cash et al. 2017a; Rodrigues et al. 2024). Here, the interaction between the newly designed intervention and the existing infrastructure (in which the intervention will be implemented) must be carefully explored, designed,

assessed, and maintained to increase intervention performance. This highlights the need to continuously refine facilitators' and users' acceptance and adherence of the intervention (Nastasi et al. 2000; Sekhon et al. 2017). In our sample, more recent publications hint at increased attention for this issue by combining intervention effectiveness measures with acceptance and adherence measures (Palsola et al. 2020; Preece et al. 2021). This has the potential to further connect to and draw on design fields, including user experience (UX), user interaction (UI), and user centric (UC) design, which have long understood that product success (including compliance and adherence) is critically linked to users' product acceptance (including likeability and meaningfulness) (Allanwood & Beare, 2014; Norman et al. 1995). Thus, combining traditional intervention implementation and effectiveness testing activities with UX, UI, and UC approaches has the potential to substantially increase intervention performance in the fuzzy back end as well as explain the success or failure of many – at first glance – apparently similar interventions. Further, this raises questions to at what degree – and if at all – intervention replication is possible. This provides the basis for three key implications for practice:

6) Interventions – both individually and collectively as a system – are implemented in **existing, complex infrastructures**. Behavioural designers must not only choose and evaluate design features towards internal synergy, but also towards synergy with (or intentional challenging of) the external contextual factors of the existing infrastructure.
7) Implementation of interventions in existing and evolving infrastructures demands dealing with a **fuzzy back end.** Behavioural designers should employ abductive exploration and creative approaches to increase intervention performance through iterative refinement of the match between interventions and existing infrastructures, whilst ensuring constant alignment between definitions of behavioural problems, behavioural goals, and intervention design.
8) Intervention performance is not only dependant on synergy between intervention and infrastructure, but also between **intervention and user**. Behavioural designers should combine more traditional late-stage development work and intervention performance testing with UX, UI, and UC approaches to iteratively refine the 'final' intervention design to increase performance via acceptance, likeability, and meaningfulness.

5.2.2. Intervention 'afterlife' and beyond the fuzzy back end

In testing intervention performance, across our sample BCTs were treated as 'the active ingredient' (e.g. Doğru et al. 2021; Jovicic et al. 2015) i.e. BCTs are understood as the primary drivers of behaviour change. Also, intervention performance measures were primarily linked to BCTs and carried out as a 'snap-shot' in stable contexts. For example, intervention performance was typically assessed via a trial with one or more of the following (i) trial outcome only (no comparison data) (e.g. Costello et al. 2018), (ii) 'usual practice' (pre–post baseline) (e.g. Pincus et al. 2015), (iii) randomised control trials (e.g. Kaur et al. 2022), and (vi) alternative intervention designs (e.g. Schroé et al. 2020). Notably, none of these were geared to account for embodied design features, and only very few considered potential interactions with context over time. Yet, embodied design features also play a critical role in

usability and likability linked to acceptance and adherence measures, which can co-develop with contextual factors over time. For example, studies show that even small differences in the wording of text messages (Garcia et al. 2019; Hallsworth et al. 2017), or the shape of otherwise identical products (Cash et al. 2017b) can steer behavioural choices. Further, from our sample, Thomas-Walters et al. (2020), as part of studying the long-term effectiveness of an intervention package, reflected that they could not tell to what degree interference from a change in the contextual infrastructure boosted the positive results they initially ascribed to their intervention package. More generally, multiple studies in our sample dealt with behaviour change connecting to habits (e.g. Liew et al. 2021; Meade et al. 2019) and beliefs or attitudes (e.g. Knight et al. 2021; Langdridge et al. 2019), which typically demand continuous reinforcement over a relatively long time span (Hong et al. 2013; Tolan et al. 2009) making them particularly sensitive to changes in contextual infrastructure (Kwasnicka et al. 2016). This is at odds with the 'static' data point provided by a trial and gives little insight into dynamic behavioural effects over time. This means that in addition to comparing underlying BCTs, it is at least equally critical to also consider and report how and to what extend the embodied intervention design features can be compared to either baselines or alternative designs, and how behaviour can be maintained over time in interplay with contextual infrastructures (Maier & Cash 2022).

In terms of accounting for embodied design features, the dominant behavioural design practice documented across our sample would benefit from explorative design activities to assess intervention effectiveness prior to pilot studies through smaller, more agile iteration of – for example – parallel small-scale experiments, A-B (pre–post) baseline testing (Doğru et al. 2021), and user experience (UX) input (Bate & Robert, 2007). Following this, it is necessary to acknowledge that intervention performance – after implementation – will inevitably change over time, due to changes in the contextual infrastructure. This demands longitudinal assessment and maintenance of intervention performance aligned with strategic choices about intervention afterlife related to business and management factors such as expected lifetime, maintenance management, redesign, up/down scaling, or intervention end of life. Here, parallels can be drawn with service and systems design (Van Halen et al. 2005; Vasantha et al. 2012) for which strategic choices and aims play a critical role in informing design during and after development. Further, both service and systems design provide frameworks for understanding and dealing with complex interactions with dynamic contextual infrastructure over time. For example, the systems engineering V-model (Childs 2013) could be integrated with current behavioural design practices to help structure testing at different levels of abstraction and acknowledge both the fuzzy front and back ends (Almqvist 2017). This provides the basis for three key implications for practice:

9) The behavioural impact of interventions is **relative**, and linked to both BCTs and embodied design features. Behavioural designers should analyse both BCTs' and embodied design features' impact on intervention performance through small-scale, iterative performance testing during intervention development prior to more resource intensive pilot studies, where behavioural effects should be measured through comparison testing.
10) Intervention performance is **dynamic**, and will change over time either due to the nature of the intervention itself, changes in the infrastructure it is implemented in,

or both. Behavioural designers should take a longitudinal, systemic approach to designing for, measuring, and maintaining intervention performance.

11) In **intervention 'afterlife'** (post implementation), both intervention relevance and performance will change over time. Behavioural designers should design for intervention management, addressing action planning and including expected intervention lifetime, maintenance, reinforcement, refinement, redesign, or shutdown. In particular, designers should consider and plan for intervention maintenance and end-of-life.

5.3. Behavioural design empathy and ethics

The final theme highlights the centrality of empathy and ethics in behavioural design (linked to findings across sections 4.1-4.6).

5.3.1. The behavioural designer's power

The decisions designers take have huge impact on both the behavioural design process and its outcomes, yet only one paper in our sample (Patton et al. 2020) explicitly reflected on and documented this. In the current literature, design related decisions are typically tacit and therefore their consequences are often unanticipated. This creates a critical challenge to the major purpose of most behavioural design work in changing behaviour for the better. Specifically, changing behaviour for the better is not straightforward, and having good intentions does not necessarily translate into a responsible intervention for individual users, society, or both (Tromp & Hekkert 2018). Without being aware of central choices, their nature, and their potential consequences it is impossible to meet the demands of transparency, explicability, and ethical responsibility integral to behavioural design (Peeters et al. 2022). Notably in our sample, this dependency is explicitly acknowledged for behavioural science related decisions but not design related ones. Hence, it is critical to acknowledge both the power of the behavioural designer as well as the relevance of the full spectrum of decisions taken.

Here, parallels can be drawn with discussions around design ethics. First, behavioural designers should not only consider when it is right or appropriate to change behaviours but also in what instances change is mandated (Tengland 2012; Tromp & Hekkert 2017), applying for example Berdichevsky & Neuenschwander's (1999) eight ethical principles that support a designer in self-reflection. In addition, behavioural designers – and other decision makers – should pay attention to how chosen embodied design features directly impact user behaviour, experience, and perceptions (Cash et al. 2017b; Tromp et al. 2011; Wilson et al. 2016). For example, the dimensions on café furniture can dictate which body types can use the café, and implementing digitalisation platforms using sensitive data between 'non-digital' citizens, e.g. elderlies, and government institutions or banks can tempt users to share pin codes with third parties to get necessary digital help. While it is impossible to account for all possible outcomes, especially behavioural designers are obliged to engage with possible ethical and equity concerns both during the design process and after intervention implementation (van den Hoven et al. 2015). Reflecting on the prior sections, ethical issues are particularly prominent in the fuzzy front and back ends as well as in the intervention afterlife. This provides the basis for two key implications for practice:

12) Design decisions are **implemented power**. Behavioural designers – and other decision makers – should acknowledge these decisions and carefully consider behavioural consequences of their design intentions and choices.
13) **Design decisions** have impact on processes and outcomes across the whole scope of behavioural design. Behavioural designers should explicitly describe and document these design decisions to systematically reflect on and match design intentions to relevant process activities, as well as anticipating and evaluating relevant outcomes and consequences.

5.3.2. The behavioural designers' responsibility

Behavioural design utilises the fact that interaction between humans and contextual infrastructures (including artefacts and other people) inevitably affects behaviour – whether intended or not (Miltenberger 2016; Tromp & Hekkert 2017). This requires a deep understanding of the target group, and its diversity in context, going beyond behaviour to include parameters like usability, likeability, meaningfulness, and appropriateness (Van Arkel & Tromp 2022; Sääksjärvi & Gonçalves 2018; Bate & Robert 2007). This is partially at odds with the practices reported in our sample, as no papers reported exploration activities connected to investigating and defining inclusion boundaries of target groups. For example, while some studies in our sample document reflections on what design features the target group might find appealing (e.g. Knight et al. 2021), and other studies reported involving stakeholders representing users, facilitators, or both (e.g. Barnett et al. 2014), both of these were primarily linked to validation activities. In general, across our sample, in scoping the design space, emphasis was first placed on theoretical desk research (e.g. Johnson & May 2015), rather than user-involvement that might confront issues of empathy and ethics early in the design process. Across our sample, the only major acknowledgement of empathy and ethics was the use of the APEASE checklist (Michie et al. 2014). This was repeatedly referred to in connection with early stage design decisions (e.g. Newton & Asimakopoulou 2018; Perski et al. 2022). The checklist includes and encourages reflection on intervention: Acceptability, Practicability, Effectiveness, Affordability, Spill-over effects, and Equity, and hence has the potential to address issues connected to 'user-intervention'-interaction, 'intervention-infrastructure'-interaction, and unintended behavioural consequences. However, even when APEASE was applied, the effectiveness assessment was solely focused on the interventions effect on the target group, ignoring the wider context exposed to the intervention. This highlights a critical misalignment between the scope for empathetic and ethical issues and current practice.

Here, behavioural designers would benefit from systematically conducting user emphasising activities or co-design (Bate & Robert 2007; Dunne & Martin 2006; Michlewski 2008) – where intervention users and facilitators take active part in designing the intervention – to critically consider empathy and ethics issues. Further, insights from such design practices reveal the importance of: (i) exploring the scope for and assessing anticipated and unanticipated effects, and (ii) paying attention to how the intervention affects behaviour across different sub-groups as well as beyond the target group. For this purpose, Emery et al. (2021) offer the INCASE framework to anticipate 6 types of unintended behavioural consequences including: Intended behaviour achieved (but the existing infrastructure cannot cope), Non-target audience exposed to intervention, Compensatory behaviours (moral licensing), Additional behaviour changes (spill-over effects), Signalling (causing unintended changed

Table 5. Theoretical implications phrased as research questions and practical recommendations for behavioural design.

	Theme	Theoretical implications for behavioural design	Practical implications for behavioural designers
1	Behavioural design problems are **wicked** (sec. 4.1.1)	• How can wicked behavioural problems be understood and treated, whilst retaining the value of existing practices geared toward tame problems?	• Behavioural designers should acknowledge that wicked problems deal with multifaceted, and intertwined potential problem causes. • Treating behavioural design problems as wicked rather that tame decreases the risk of solving 'wrong problems', solving them in 'wrong ways', or both.
2	Wicked problems demand **fuzzy front end** approaches (sec. 4.1.1)	• How can abductive exploratory approaches be effectively integrated with more deductive working in early stage scoping and problem refinement? • How should traditional fuzzy front end practices be adapted to the particular uncertainties associated with behavioural problems?	• Behavioural designers should iteratively explore and reframe co-evolved behavioural problems and solutions prior to deductive development work. • Applying abductive approaches prior to deductive development work decreases risks and resource demands.
3	Iterative reframing should be **documented** (sec. 4.1.1)	• How can design work and decisions be understood and effectively documented in a comparable manner to more deductive work? • How can reframing activities and outcomes be more systematically understood and documented?	• Behavioural designers should document fuzzy front end decisions and co-evolution of problem/solution. • Both small and large scale (re)framing have significant impact on the project process and outcomes and should be documented.
4	Interventions reflect **complex combinations** of BCTs and design features (sec. 4.1.2)	• How can combinations of BCTs and embodied design features be collectively understood and analysed? • How can stakeholder involvement be more effectively used to help navigate in the high uncertainty connected to the many possible options for embodied design features?	• Behavioural designers should think of embodied design features as BCT facilitators, thus equally important as BCTs in realising behavioural outcomes. • Embodied design features should be developed via abductive exploration and creative techniques, prior to resource intensive deductive development work.
5	Interventions are **complex and comprehensive** (sec. 4.1.2)	• How can complex interventions combining multiple, intertwined elements acting as an intervention system, be understood and effectively documented? • How can internal systemic interactions be mapped and worked with during behavioural design?	• Behavioural designers should explore how each intervention element will affect the entire system. • Deploying exploratory prototyping and similar activities should be used to unpack possible interactions prior to committing to a final design.
6	Interventions in **existing, complex infrastructures** (sec. 4.2.1)	• How can interaction between the internal intervention system and external contextual infrastructure be understood and documented? • How can systemic effects be accounted for in the design process?	• Behavioural designers should not only evaluate design features for internal synergy, but also for synergy with (or intended challenge of) the externally existing infrastructure.

(*continued*).

Table 5. Continued.

	Theme	Theoretical implications for behavioural design	Practical implications for behavioural designers
7	Development of synergy demands **fuzzy front end** approaches (sec. 4.2.1)	• How can abductive exploratory and creative approaches be effectively integrated with more deductive working in late-stage behavioural design? • How should traditional fuzzy back end practices be adapted to the particular uncertainties associated with behavioural solutions?	• Behavioural designers should increase intervention performance whilst ensuring alignment between defined behavioural problems, goals, and intervention design, by iteratively refining the match between interventions and existing infrastructures through co-evolutionary explorative and creative approaches.
8	Interventions must make sense for **users** (sec. 4.2.1)	• How can synergy between intervention and infrastructure, and intervention and user be understood and designed for? • How should traditional systemic design practices be adapted to behavioural design?	• Behavioural designers should combine traditional late-stage development and performance testing with UX, UI, and UC approaches to iteratively refine the intervention design to increase performance via acceptance, likeability, and meaningfulness.
9	Behavioural impact is **relative** (sec. 4.2.2)	• How can behavioural design shift from assessing intervention performance linked primarily to BCTs, to impact accounting for BCTs and embodied design features in conjunction?	• Behavioural designers should analyse both BCTs' and embodied design features' impact on intervention performance through small-scale, iterative testing during intervention development.
10	Intervention performance is **dynamic** (sec. 4.2.2)	• How can a longitudinal, systemic approach to designing for, measuring, and maintaining of intervention performance be accommodated in necessarily time limited behavioural design projects?	• Behavioural designers should design for, measure, and document intervention performance over time. • Longitudinal performance assessment will inform relevant action during the intervention afterlife.
11	Interventions have an **afterlife** (sec. 4.2.2)	• How can long term testing and effect assessments be included in behavioural design approaches? • How can the complexities of intervention afterlife and end of life be understood and managed?	• Behavioural designers should design for intervention management addressing action planning, including expected lifetime, maintenance, reinforcement, refinement, redesign, and end of life.
12	Design decisions equal **power** (sec. 4.3.1)	• How can design decisions be evaluated in context alongside more deductive decision making? • How can design consequences be evaluated and accounted for during the design process?	• Behavioural designers – and other decision makers – should acknowledge the power of design decisions considering and reflecting on the consequences of their design intentions and choices.
13	**Decisions** impact process and outcomes (sec. 4.3.1)	• How can design decisions' impact on processes and outcomes across the whole scope of behavioural design projects be understood and effectively managed?	• Behavioural designers should explicitly describe design decisions to systematically reflect on alignment between intentions, process activities, and outcomes.
14	Interventions must be **responsibly** designed (sec. 4.3.2)	• How can primarily theory informed decision making be combined with stakeholder informed decision making and other empathetic practices? • How can empathetic and ethical considerations be more systematically integrated in the design process?	• Behavioural designers must engage in co-design with stakeholders to understand their needs and desires to deliver an empathetic, ethical, and responsible match between intervention and user.

(*continued*).

Table 5. Continued.

	Theme	Theoretical implications for behavioural design	Practical implications for behavioural designers
15	Interventions have **various impact** (sec. 4.3.2)	• How can behavioural design shift from primarily focusing on predictable outcomes of target groups, to include traditionally unanticipated outcomes, e.g. behavioural changes in non-target groups?	• Behavioural designers should apply reflective approaches to describe and respond to anticipated and unanticipated behavioural consequences across both target and non-target groups, and over time.

attitudes of recipients), and Emotional Impact (unintendedly reinforcing existing, problematic behavioural patterns). Unanticipated and unintended consequences is a risk both in terms of miss-concluding why the intervention performs as measured in effectiveness assessment (Stegenga 2015), as well as introducing new behaviours that either neutralise or worsen the starting point of the intended behavioural change (Maier & Cash 2022). As such, behavioural designers must not only pay careful attention to behavioural consequences (Barari et al. 2020) in the form of behavioural goals (anticipated and desirable), but also to trade-offs (anticipated and undesirable), serendipities (unanticipated and desirable), and classic negative unintended consequences (unanticipated and undesirable) (Emery et al. 2021). In accommodating for this inherent uncertainty, behavioural designers would benefit from engaging with relevant stakeholders to inform the up-front need to work with ethical issues across the whole life cycle of an intervention. Here, UNICEF has successfully employed interactive, co-design strategies ensuring a holistic view of people's needs, desires, as well as barriers and facilitators to change, by applying user centred (UC) approaches including relevant stakeholders across intervention development, prototyping, and implementation. This provides the basis for two key implications for practice:

14) Interventions must **make sense** and be **responsibly** designed for users as well as other contextually relevant stakeholders. Behavioural designers must engage with stakeholders to understand their needs and desires in order to deliver an empathetic, ethical, and responsible match between intervention and user.
15) Interventions have **various impacts** on both target and non-target groups, which are only partially predictable. Behavioural designers should apply reflective approaches to describe, consider, and respond to behavioural consequences of interventions both in terms of anticipated and unanticipated effects across target and non-target groups over time.

5.4. A way forward for behavioural design

We bring the insights and implications discussed above together by phrasing them as research questions and practical recommendations for behavioural design in Table 5. This serves to point a way forward for behavioural design research and practice.

In addition, we propose a reconceptualization of prior, predominantly linear behavioural design process models (Aunger & Curtis 2016; Cash et al. 2017a; Fogg 2009; Michie et al. 2014; Skivington et al. 2021; Wendel 2013). The proposed Behavioural Design IM-PACT process model (Figure 3) stands for: Integrated Methodology – People (*users and other*

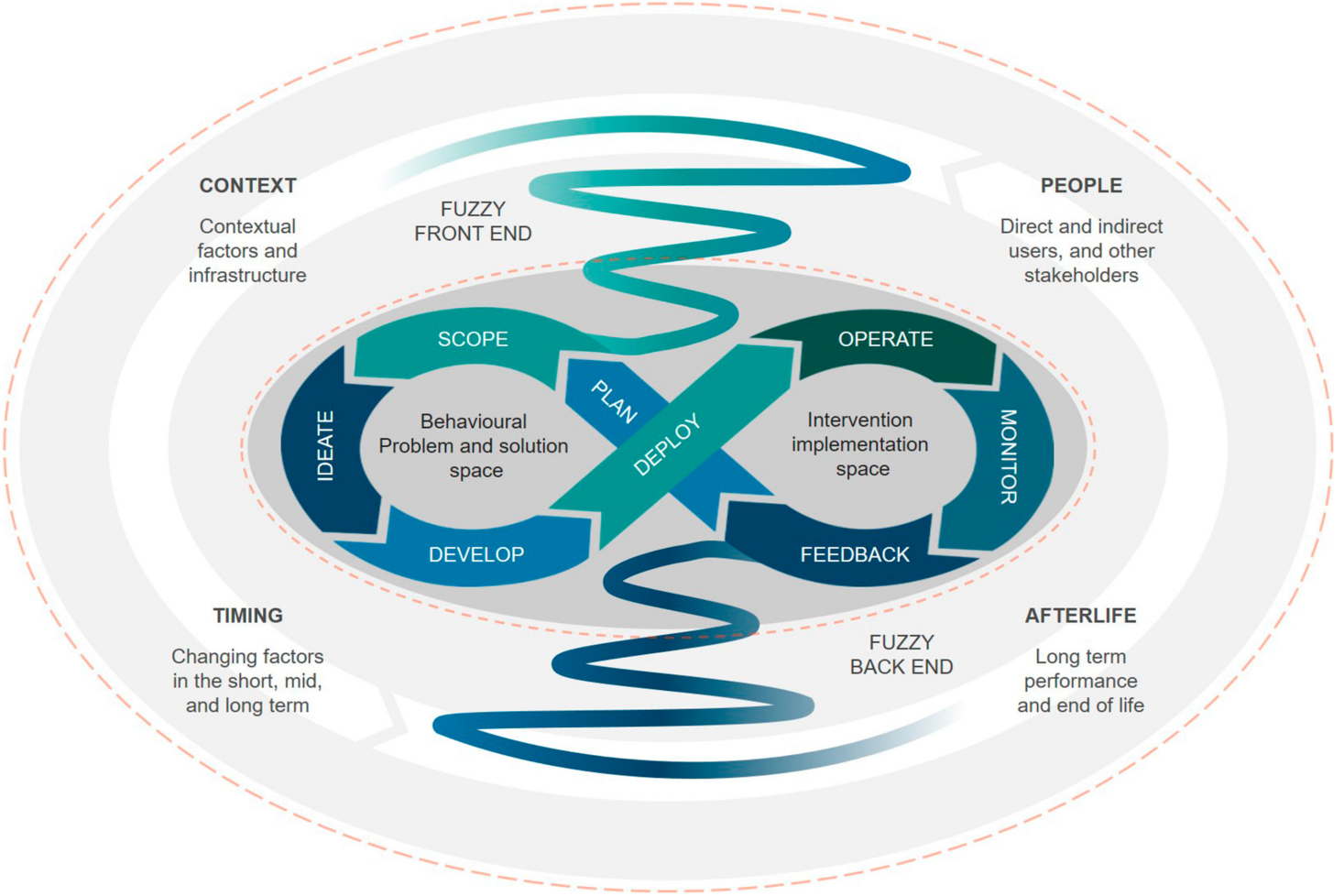


Figure 3. The Behavioural Design IM-PACT process model. Reconceptualising behavioural design as a flexible, iterative double-loop development process, bringing together the behavioural problem/solution and intervention implementation spaces, as well as showcasing the inevitable co-evolutionary interaction between surrounding PACT (People, Afterlife, Timing, Context) factors.

stakeholders), Afterlife (*long term performance and end of life*), Context (*contextual factors and infrastructure*), and Timing (*short, middle and long term changing factors*).

In the IM-PACT process model we elucidate how behavioural design can (but does not necessarily) emerge from an existing dynamic context via a fuzzy front end, develop iteratively via a double-loop connecting the behavioural problem/solution space and the intervention implementation space, and feed out via a fuzzy back end. In addition, we illustrate how this double-loop process is both affected by PACT (People, Afterlife, Context, Timing) factors during development and implementation, as well as affects those same PACT factors in implementation. This drives a co-evolutionary interaction between implemented intervention and PACT factors, which sets the stage for subsequent development processes. Thus, the Behavioural Design IM-PACT process model showcases how design decisions are informed by and feed back into a system between the existing fuzzy front end PACT factors, the intervention development itself, and the 'updated' PACT factors after implementing the developed intervention in the fuzzy (also called 'messy') back end. Lastly, the double-loop development process allows for flexible development practices, accounting for the inevitable variation of needed iteration, number of cycles in each space, PACT factors, and project specific starting points.

Based on the characteristics illustrated in Figure 3, the Behavioural Design IM-PACT process model offers a concrete means of shifting from behavioural design methodology primarily rooted in either design or behavioural science to an integrative, flexible approach.

6. Conclusion

In this paper, we – for the first time – examined behavioural design decision making and methodology through a systematic literature review of 57 behavioural design papers. Our analysis based on the Architecture of Design Doing (ADD) framework revealed major discrepancies in the application and reporting of methodological elements. Our key findings, and concrete examples of these discrepancies are summarised in Table 4, and formed the basis for identifying three major themes that challenge current understanding of behavioural design: *Complex behavioural design space*, *Systemic behavioural design*, and *Behavioural design empathy and ethics*. The discussion of themes gave rise to numerous implications phrased as research questions and recommendations for practice presented in Table 5, as well as directed a fundamental reconceptualization of how we think about and approach behavioural design. Figure 3 presents our proposed Behavioural Design IM-PACT process model. This integrates a fuzzy front and back end, feeding into/out of an iterative double-loop development process (connecting the behavioural problem/solution space and implementation space), all shaped by a co-evolutionary interaction with the surrounding 'PACT' factors of: People (*users and other stakeholders*), Afterlife (*long term performance and end of life*), Context (*contextual factors and infrastructure*), and Timing (*short, middle and long term changing factors*).

In conclusion, we highlight that acknowledging and working with design approaches in a truly integrative manner is not only essential to effective and successful behavioural design, but also to ethical and empathetic processes and outcomes. Our work points towards a more cohesive, flexible, and integrative behavioural design practice, that leverages the best of both design and behavioural science, with far reaching implications for researchers across fields.

Acknowledgements

The authors would like to thank the three research assistants, Emil Folmer, Frederik Høgdal, and Rikke Straarup, for their dedicated contribution to the literature review.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Camilla K. E. Bay Brix Nielsen http://orcid.org/0000-0001-5718-8241
Philip Cash http://orcid.org/0000-0001-6498-0237
Jaap Daalhuizen http://orcid.org/0000-0003-3122-8086

References

Adams, R. S., and C. J. Atman. 1999. Cognitive Processes in Iterative Design Behavior. In FIE'99 Frontiers in Education. 29th Annual Frontiers in Education Conference. Designing the Future of Science and Engineering Education. Conference Proceedings (IEEE Cat. No. 99CH37011 (Vol. 1, pp. 11A6-13). IEEE.

Alexander, C. 1971. Preface to the Paperback Edn of Notes on the Synthesis of Form.

Allanwood, G., and P. Beare. 2014. *Basics Interactive Design: User Experience Design: Creating Designs Users Really Love*. London. UK: A&C Black.

Almqvist, F. 2017. "The Fuzzy Front-end and the Forgotten Back-end: User Involvement in Later Development Phases." *The Design Journal* 20 (sup1): S2524–S2533. https://doi.org/10.1080/14606925.2017.1352765.

Andrade, A. Q., V. T. Leblanc, L. M. Kalisch-Ellett, N. L. Pratt, A. Moffat, N. Blacker, … E. E. Roughead. 2020. "Determinants of Usefulness in Professional Behaviour Change Interventions: Observational Study of a 15-Year National Program." *BMJ Open* 10: e038016. https://doi.org/10.1136/bmjopen-2020-038016.

Andreasen, M. M., S. Kähler, K. G. Swift, and T. Lund. 1988. *Design for Assembly*. Heidelberg, Germany: Ifs.

Atkins, L., A. Sallis, T. Chadborn, K. Shaw, A. Schneider, S. Hopkins, … F. Lorencatto. 2020. "Reducing Catheter-Associated Urinary Tract Infections: A Systematic Review of Barriers and Facilitators and Strategic Behavioural Analysis of Interventions." *Implementation Science* 15 (1): 1–22. https://doi.org/10.1186/s13012-020-01001-2.

Aunger, R., and V. Curtis. 2016. "Behaviour Centred Design: Towards an Applied Science of Behaviour Change." *Health Psychology Review* 10 (4): 425–446. https://doi.org/10.1080/17437199.2016.1219673.

Band, Rebecca, Katherine Bradbury, Katherine Morton, Carl May, Susan Michie, Frances S. Mair, Elizabeth Murray, Richard J. McManus, Paul Little, and Lucy Yardley. 2017. "Intervention Planning for a Digital Intervention for Self-Management of Hypertension: A Theory-, Evidence- and Person-Based Approach." *Implementation Science* 12 (1): 1–13. https://doi.org/10.1186/s13012-017-0553-4.

Barari, S., S. Caria, A. Davola, P. Falco, T. Fetzer, S. Fiorin, … F. R. Slepoi. 2020. Evaluating COVID-19 Public Health Messaging in Italy: Self-Reported Compliance and Growing Mental Health Concerns. MedRxiv.

Barnett, K. N., M. Bennie, S. Treweek, C. Robertson, D. J. Petrie, L. D. Ritchie, and B. Guthrie. 2014. "Effective Feedback to Improve Primary Care Prescribing Safety (EFIPPS) a Pragmatic Three-arm Cluster Randomised Trial: Designing the Intervention (ClinicalTrials.gov Registration NCT01602705)." *Implementation Science* 9: 133. https://doi.org/10.1186/s13012-014-0133-9.

Bate, P., and G. Robert. 2007. *Bringing User Experience to Healthcare Improvement: The Concepts, Methods and Practices of Experience Based Design*. Oxford: Radcliffe.

Beall, R. F., N. Baskerville, M. Golfam, S. Saeed, and J. Little. 2014. "Modes of Delivery in Preventive Intervention Studies: A Rapid Review." *European Journal of Clinical Investigation* 44 (7): 688–696. https://doi.org/10.1111/eci.12279.

Bell, S., H. S. Mishra, L. R. Elliott, R. Shellock, P. Vassiljev, M. Porter, … M. P. White. 2020. "Urban Blue Acupuncture: A Protocol for Evaluating a Complex Landscape Design Intervention to Improve Health and Wellbeing in a Coastal Community." *Sustainability* 12 (10): 4084. https://doi.org/10.3390/SU12104084.

Bentley, K. H., E. M. Kleiman, G. Elliott, J. C. Huffman, and M. K. Nock. 2019. "Real-time Monitoring Technology in Single-Case Experimental Design Research: Opportunities and Challenges." *Behaviour Research and Therapy* 117: 87–96. https://doi.org/10.1016/j.brat.2018.11.017.

Berdichevsky, D., and E. Neuenschwander. 1999. "Toward an Ethics of Persuasive Technology." *Communications of the ACM* 42 (5): 51–58. https://doi.org/10.1145/301353.301410.

Blackhurst, J., T. Wu, and P. O'grady. 2004. "Network-Based Approach to Modelling Uncertainty in a Supply Chain." *International Journal of Production Research* 42 (8): 1639–1658. https://doi.org/10.1080/00207540303600016460 64.

Boland, R., and F. Collopy, eds. 2004. *Managing as Designing*. Stanford, CA: Stanford Business Books, 298.

Bondaronek, P., A. Slee, F. L. Hamilton, and E. Murray. 2019. "Relationship Between Popularity and the Likely Efficacy: An Observational Study Based on a Random Selection on top-Ranked Physical Activity Apps." *BMJ Open* 9 (11): e027536–9. https://doi.org/10.1136/bmjopen-2018-027536.

Borek, A. J., C. Abraham, J. R. Smith, C. J. Greaves, and M. Tarrant. 2015. "A Checklist to Improve Reporting of Group-Based Behaviour-Change Interventions Biostatistics and Methods." *BMC Public Health* 15 (1): 1–11. https://doi.org/10.1186/s12889-015-2300-6.

Bowers, H. M., T. Kendrick, M. Glowacka, S. Williams, G. Leydon, C. May, … A. W. A. Geraghty. 2020. "Supporting Antidepressant Discontinuation: The Development and Optimisation of a Digital Intervention for Patients in UK Primary Care Using a Theory, Evidence and Person-Based Approach." *BMJ Open* 10 (3): e03231. https://doi.org/10.1136/bmjopen-2019-032312.

Bryan, C. J., E. Tipton, and D. S. Yeager. 2021. "Behavioural Science is Unlikely to Change the World Without a Heterogeneity Revolution." *Nature Human Behaviour* 5: 980–989. https://doi.org/10.1038/s41562-021-01143-3.

Buchanan, R. 1992. "Wicked Problems in Design Thinking." *Design Issues* 8 (2): 5–21. https://doi.org/10.2307/1511637.

Bucher, A. 2020. *Engaged: Designing for Behavior Change*. New York, USA: Rosenfeld Media.

Cash, P., X. V. Gamundi, I. Echstrøm, and J. Daalhuizen. 2022. "Method Use in Behavioural Design: What, How, and Why?" *International Journal of Design* 16 (1): ISSN 1991-3761.

Cash, P. J., C. G. Hartlev, and C. B. Durazo. 2017a. "Behavioural Design: A Process for Integrating Behaviour Change and Design." *Design Studies* 48: 96–128. https://doi.org/10.1016/j.destud.2016.10.001.

Cash, P., C. Holm-Hansen, S. B. Olsen, M. L. Christensen, and Y. M. T. Trinh. 2017b. "Uniting Individual and Collective Concerns Through Design: Priming Across the Senses." *Design Studies* 49: 32–65. https://doi.org/10.1016/j.destud.2017.01.002.

Cash, P., P. Khadilkar, J. Jensen, C. Dusterdich, and R. Mugge. 2020. "Designing Behaviour Change: A Behavioural Problem/Solution (BPS) Matrix." *International Journal of Design* 14 (2): 65–83.

Cash, P., A. Wrobel, A. Maier, and J. P. Hansen. 2023. "Understanding Long-Term Behaviour Change Techniques: A Mixed Methods Study." *Journal of Engineering Design* 34: 1–28.

Chang, M., and C. J. Brainerd. 2022. "Association and Dissociation Between Judgments of Learning and Memory: A Meta-Analysis of the Font Size Effect." *Metacognition and Learning* 17 (2): 443–476. https://doi.org/10.1007/s11409-021-09287-3.

Childs, P. R. 2013. *Mechanical Design Engineering Handbook*. Oxford, UK: Butterworth-Heinemann.

Costello, N., J. McKenna, L. Sutton, K. Deighton, and B. Jones. 2018. "Using Contemporary Behavior Change Science to Design and Implement an Effective Nutritional Intervention Within Professional Rugby League." *International Journal of Sport Nutrition and Exercise Metabolism* 28 (5): 553–557. https://doi.org/10.1123/ijsnem.2017-0298.

Cross, N. 2006. "Understanding Design Cognition." In *Designerly Ways of Knowing*, 77–93. London: Springer publishers.

Daalhuizen, J., R. Timmer, M. van der Welie, and P. Gardien. 2019. "An Architecture of Design Doing: A Framework for Capturing the Ever-Evolving Practice of Design to Drive Organizational Learning." *International Journal of Design* 13 (1): 37–52.

der Wardt, V. van, C. di Lorito, and A. Viniol. 2021. "Promoting Physical Activity in Primary Care: A Systematic Review and Meta-Analysis." *British Journal of General Practice* 71 (706): E399–E405. https://doi.org/10.3399/BJGP.2020.0817.

Desveaux, L., K. McBrien, L. Barnieh, and N. M. Ivers. 2019. "Mapping Variation in Intervention Design: A Systematic Review to Develop a Program Theory for Patient Navigator Programs Medical and Health Sciences Public Health and Health Services." *Systematic Reviews* 8 (1): 1–14. https://doi.org/10.1186/s13643-018-0920-5.

Doğru, O. C., T. L. Webb, and P. Norman. 2021. "What is the Best way to Promote Cycling? A Systematic Review and Meta-Analysis." *Transportation Research Part F: Traffic Psychology and Behaviour* 81 (January): 144–157. https://doi.org/10.1016/j.trf.2021.06.002.

Dorst, K. 2006. "Design Problems and Design Paradoxes." *Design Issues* 22 (3): 4–17. https://doi.org/10.1162/desi.2006.22.3.4.

Dorst, K., and N. Cross. 2001. "Creativity in the Design Process: Co-Evolution of Problem Solution." *Design Studies* 22: 425–437. https://doi.org/10.1016/S0142-694X(01)00009-6.

Dunn, G., R. Emsley, H. Liu, S. Landau, J. Green, I. White, and A. Pickles. 2015. "Evaluation and Validation of Social and Psychological Markers in Randomised Trials of Complex Interventions in Mental Health: A Methodological Research Programme." *Health Technology Assessment* 19 (93): 1–116. https://doi.org/10.3310/hta19930.

Dunne, D., and R. Martin. 2006. "Design Thinking and how it Will Change Management Education: An Interview and Discussion." *Academy of Management Learning & Education* 5 (4): 512–523. https://doi.org/10.5465/amle.2006.23473212.

Dyson, J., and F. Cowdell. 2021. "How is the Theoretical Domains Framework Applied in Designing Interventions to Support Healthcare Practitioner Behaviour Change? A Systematic Review." *International Journal for Quality in Health Care* 33 (3): 1–15. https://doi.org/10.1093/intqhc/mzab106.

Emery, A., et al. 2021. https://gcs.civilservice.gov.uk/wp-content/uploads/2021/07/GCS_IN_CASE_A_behavioural_approach_to_anticipating_unintended_consequences.pdf.

Fabio, R. A., and A. Croce. 2024. "The Role of Peace Attitudes on Sustainable Behaviors: An Exploratory Study." *Behavioral Sciences* 14 (2): 120. https://doi.org/10.3390/bs14020120.

Fogg, B. J. 2009. "A Behavior Model for Persuasive Design." In *Proceedings of the 4th International Conference on Persuasive Technology*, 1–7.

French, D. P., E. K. Olander, A. Chisholm, and J. Mc Sharry. 2014. "Which Behaviour Change Techniques are Most Effective at Increasing Older Adults' Self-Efficacy and Physical Activity Behaviour? A Systematic Review." *Annals of Behavioral Medicine* 48 (2): 225–234. https://doi.org/10.1007/s12160-014-9593-z.

Garcia, S. M., et al. 2019. https://www.bi.team/wp-content/uploads/2019/11/Guatemala-6-TR-Encourage-timely-declaration-of-tax-1.pdf.

Govender, R., C. E. Wood, S. A. Taylor, C. H. Smith, H. Barratt, and B. Gardner. 2017. "Patient Experiences of Swallowing Exercises After Head and Neck Cancer: A Qualitative Study Examining Barriers and Facilitators Using Behaviour Change Theory." *Dysphagia* 32 (4): 559–569. https://doi.org/10.1007/s00455-017-9799-x.

Goyal, S., P. Morita, G. F. Lewis, C. Yu, E. Seto, and J. A. Cafazzo. 2016. "The Systematic Design of a Behavioural Mobile Health Application for the Self-Management of Type 2 Diabetes." *Canadian Journal of Diabetes* 40 (1): 95–104. https://doi.org/10.1016/j.jcjd.2015.06.007.

Grudniewicz, A., R. Kealy, R. N. Rodseth, J. Hamid, D. Rudoler, and S. E. Straus. 2015. "Economic Evaluation of the Practical Approach to Lung Health and Informal Provider Interventions for Improving the Detection of Tuberculosis and Chronic Airways Disease at Primary Care Level in Malawi: Study Protocol for Cost-Effectiveness Analysis." *Implementation Science* 10: 1–12. https://doi.org/10.1186/s13012-014-0195-8.

Günther, J., and K. Ehrlenspiel. 1999. "Comparing Designers from Practice and Designers with Systematic Design Education." *Design Studies* 20 (5): 439–451. https://doi.org/10.1016/S0142-694X(99)00019-8.

Hagger, M. S., and M. Weed. 2019. "DEBATE: Do Interventions Based on Behavioral Theory Work in the Real World?" *International Journal of Behavioral Nutrition and Physical Activity* 16 (1): 1–10. https://doi.org/10.1186/s12966-019-0795-4.

Hallsworth, M., J. A. List, R. D. Metcalfe, and I. Vlaev. 2017. "The Behavioralist as tax Collector: Using Natural Field Experiments to Enhance tax Compliance." *Journal of Public Economics* 148: 14–31. https://doi.org/10.1016/j.jpubeco.2017.02.003.

Hansen, P. G. 2018. "What are We Forgetting?" *Behavioural Public Policy* 2 (2): 190–197. https://doi.org/10.1017/bpp.2018.13.

Hargreaves, D., E. Mates, P. Menon, H. Alderman, D. Devakumar, W. Fawzi, … G. C. Patton. 2022. "Strategies and Interventions for Healthy Adolescent Growth, Nutrition, and Development." *The Lancet* 399 (10320): 198–210. https://doi.org/10.1016/S0140-6736(21)01593-2.

Harries, T., and R. Rettie. 2016. "Walking as a Social Practice: Dispersed Walking and the Organisation of Everyday Practices." *Sociology of Health & Illness* 38 (6): 874–883. https://doi.org/10.1111/1467-9566.12406.

Hay, L., A. H. Duffy, C. McTeague, L. M. Pidgeon, T. Vuletic, and M. Grealy. 2017. "A Systematic Review of Protocol Studies on Conceptual Design Cognition: Design as Search and Exploration." *Design Science* 3: e10. https://doi.org/10.1017/dsj.2017.11.

Heino, M. T. J., K. Knittle, C. Noone, F. Hasselman, and N. Hankonen. 2021. "Studying Behaviour Change Mechanisms Under Complexity." *Behavioral Sciences* 11 (5): 77–22. https://doi.org/10.3390/bs11050077.

Hoffmann, T. C., P. P. Glasziou, I. Boutron, R. Milne, R. Perera, D. Moher, … S. Michie. 2014. "Better Reporting of Interventions: Template for Intervention Description and Replication (TIDieR) Checklist and Guide." *Bmj* 348: g1687–g1687. https://doi.org/10.1136/bmj.g1687.

Hong, O., D. L. Chin, L. A. Fiola, and A. S. Kazanis. 2013. "The Effect of a Booster Intervention to Promote Hearing Protection Behavior in Operating Engineers." *American Journal of Industrial Medicine* 56 (2): 258–266. https://doi.org/10.1002/ajim.22091.

Hughes, C., and C. Ryan. 2024. "Can we do Better? Sustainability and Efficiency in Intervention Development and Implementation." *International Journal of Pharmacy Practice* xx: 1–2. https://doi.org/10.1093/ijpp/riae004.

Hulscher, M. E. J. L., and J. M. Prins. 2017. "Antibiotic Stewardship: Does it Work in Hospital Practice? A Review of the Evidence Base." *Clinical Microbiology and Infection* 23 (11): 799–805. https://doi.org/10.1016/j.cmi.2017.07.017.

Jalil, N. A., R. M. Yunus, and N. S. Said. 2012. "Environmental Colour Impact upon Human Behaviour: A Review." *Procedia - Social and Behavioral Sciences* 35: 54–62. https://doi.org/10.1016/j.sbspro.2012.02.062.

Jiao, R., L. Luo, J. Malmqvist, and J. Summers. 2022. "New Design: Opportunities for Engineering Design in an Era of Digital Transformation." *Journal of Engineering Design* 33 (10): 1–5. https://doi.org/10.1080/09544828.2022.2147270.

Jiao, Roger J., Feng Zhou, and Chih-Hsing Chu. 2017. "Decision Theoretic Modeling of Affective and Cognitive Needs for Product Experience Engineering: Key Issues and a Conceptual Framework." *Journal of Intelligent Manufacturing* 28 (7): 1755–1767. https://doi.org/10.1007/s10845-016-1240-z.

Johnson, M. J., and C. R. May. 2015. "Promoting Professional Behaviour Change in Healthcare: What Interventions Work, and Why? A Theory-Led Overview of Systematic Reviews." *BMJ Open* 5 (9): e008592. https://doi.org/10.1136/bmjopen-2015-008592.

Jovicic, A., B. Gardner, C. Belk, K. Kharicha, S. Iliffe, J. Manthorpe, … K. Walters. 2015. "Identifying the Content of Home-Based Health Behaviour Change Interventions for Frail Older People: A Systematic Review Protocol." *Systematic Reviews* 4 (1): 1–8. https://doi.org/10.1186/s13643-015-0138-8.

Kassavou, A., and S. Sutton. 2018. "Automated Telecommunication Interventions to Promote Adherence to Cardio-Metabolic Medications: Meta-Analysis of Effectiveness and Meta-Regression of Behaviour Change Techniques." *Health Psychology Review* 12 (1): 25–42. https://doi.org/10.1080/17437199.2017.1365617.

Kaur, S., M. Kaur, and R. Kumar. 2022. "Health Promotion Intervention to Prevent Risk Factors of Chronic Diseases: Protocol for a Cluster Randomized Controlled Trial among Adolescents in School Settings of Chandigarh (India)." *PLoS One* 17: 1–18. https://doi.org/10.1371/journal.pone.0263584.

Keyworth, C., T. Epton, J. Goldthorpe, R. Calam, and C. J. Armitage. 2020. "Acceptability, Reliability, and Validity of a Brief Measure of Capabilities, Opportunities, and Motivations ("COM-B")." *British Journal of Health Psychology* 25 (3): 474–501. https://doi.org/10.1111/bjhp.12417.

Khadilkar, P. R., and P. Cash. 2020. "Understanding Behavioural Design: Barriers and Enablers." *Journal of Engineering Design* 31 (10): 508–529. https://doi.org/10.1080/09544828.2020.1836611.

Kimbell, L., and P. E. Street. 2009. "Beyond Design Thinking: Design-as-Practice and Designs-in-Practice." In *CRESC Conference, Manchester*, 1–15.

Klusmann, V., A. J. Gow, P. Robert, and G. Oettingen. 2021. "Using Theories of Behavior Change to Develop Interventions for Healthy Aging." *Journals of Gerontology - Series B Psychological Sciences and Social Sciences* 76: S191–S205. https://doi.org/10.1093/geronb/gbab111.

Knight, H., R. Jia, K. Ayling, K. Bradbury, K. Baker, T. Chalder, J. R. Morling, et al. 2021. "Understanding and Addressing Vaccine Hesitancy in the Context of COVID-19: Development of a Digital Intervention." *Public Health* 201: 98–107. https://doi.org/10.1016/j.puhe.2021.10.006.

Kok, Gerjo, Nell H. Gottlieb, Gjalt-Jorn Y. Peters, Patricia Dolan Mullen, Guy S. Parcel, Robert A.C. Ruiter, María E. Fernández, Christine Markham, and L. Kay Bartholomew. 2016. "A Taxonomy of Behaviour Change Methods: An Intervention Mapping Approach." *Health Psychology Review* 10 (3): 297–312. https://doi.org/10.1080/17437199.2015.1077155.

Koohsari, Mohammad Javad, Suzanne Mavoa, Karen Villanueva, Takemi Sugiyama, Hannah Badland, Andrew T. Kaczynski, Neville Owen, and Billie Giles-Corti. 2015. "Public Open Space, Physical Activity, Urban Design and Public Health: Concepts, Methods and Research Agenda." *Health & Place* 33: 75–82. https://doi.org/10.1016/j.healthplace.2015.02.009.

Kunrath, K., P. Cash, and M. Kleinsmann. 2020. "Designers' Professional Identity: Personal Attributes and Design Skills." *Journal of Engineering Design* 31 (6): 297–330. https://doi.org/10.1080/09544828.2020.1743244.

Kwasnicka, D., S. U. Dombrowski, M. White, and F. Sniehotta. 2016. "Theoretical Explanations for Maintenance of Behaviour Change: A Systematic Review of Behaviour Theories." *Health Psychology Review* 10 (3): 277–296. https://doi.org/10.1080/17437199.2016.1151372.

Lambert, J. D., C. J. Greaves, P. Farrand, R. Cross, A. M. Haase, and A. H. Taylor. 2017. "Assessment of Fidelity in Individual Level Behaviour Change Interventions Promoting Physical Activity among Adults: A Systematic Review." *BMC Public Health* 17 (1): 1–13. https://doi.org/10.1186/s12889-017-4778-6.

Langdridge, Darren, Mark Davis, Lucyna Gozdzielewska, Joanna McParland, Lynn Williams, Mairi Young, Fraser Smith, Jennifer MacDonald, Lesley Price, and Paul Flowers. 2019. "A Visual Affective Analysis of Mass Media Interventions to Increase Antimicrobial Stewardship Amongst the Public." *British Journal of Health Psychology* 24 (1): 66–87. https://doi.org/10.1111/bjhp.12339.

Langley, T., D. Gillespie, S. Lewis, K. Eminson, A. Brennan, G. Docherty, and B. Young. 2021. "Developing Logic Models to Inform Public Health Policy Outcome Evaluation: An Example from Tobacco Control." *Journal of Public Health (United Kingdom)* 43 (3): 639–646. https://doi.org/10.1093/pubmed/fdaa032.

Lawson, B. 2006. *How Designers Think*. 4th edition. Oxford: Elsevier.

Lee, J. Q., K. Ying, P. Lun, K. T. Tan, W. Ang, Y. Munro, and Y. Y. Ding. 2020. "Intervention Elements to Reduce Inappropriate Prescribing for Older Adults with Multimorbidity Receiving Outpatient Care: A Scoping Review." *BMJ Open* 10 (8): e039543–10. https://doi.org/10.1136/bmjopen-2020-039543.

Lewrick, M., P. Link, and L. Leifer. 2018. *The Design Thinking Playbook: Mindful Digital Transformation of Teams, Products, Services, Businesses and Ecosystems*. Hoboken, NJ: John Wiley & Sons.

Liew, ZiHeng, Bruno Mazuquin, David R. Ellard, Eleni Karasouli, Stephen Drew, Chetan Modi, Howard Bush, Martin Underwood, and Rebecca S. Kearney. 2021. "Development of a single-session physiotherapy and self-management intervention for the treatment of primary traumatic anterior shoulder dislocation for the 'Acute Rehabilitation following Traumatic anterior shoulder dISlocAtioN (ARTISAN)' multi centre RCT." *Physiotherapy* 113: 80–87. https://doi.org/10.1016/j.physio.2021.06.002.

Lin, J. W., and L. J. Mai. 2018. "Impact of Mindfulness Meditation Intervention on Academic Performance." *Innovations in Education and Teaching International* 55 (3): 366–375. https://doi.org/10.1080/14703297.2016.1231617.

Lockton, D. 2012. Cognitive Biases, Heuristics and Decision-Making in Design for Behaviour Change. *Heuristics and Decision-Making in Design for Behaviour Change*.

Losinski, M., J. W. Maag, A. Katsiyannis, and R. P. Ennis. 2014. "Examining the Effects and Quality of Interventions Based on the Assessment of Contextual Variables: A Meta-Analysis." *Exceptional Children* 80 (4): 407–422. https://doi.org/10.1177/0014402914527243.

Maier, A., and P. Cash. 2022. "Designing for Human Behaviour in a Systemic World." In *Handbook of Engineering Systems Design*, edited by A. Maier, J. Oehmen, and P. E. Vermaas, 1–34. Cham: Springer International Publishing.

Martin, B., and B. Hanington. 2012. *Universal Methods of Design*. Beverly, MA: Rockport Publishers.

Masrur, M., J. Richardson, A. Tang, L. Thabane, and E. Cino. 2020. "Intervention-related Factors Associated with Physical Activity Maintenance among Post-Stroke Patients: A Protocol for a Systematic Review with Meta-Analysis and Meta-Regression." *JBI Evidence Synthesis* 18 (8): 1738–1750. https://doi.org/10.11124/JBISRIR-D-19-00264.

Meade, L. B., L. M. Bearne, L. H. Sweeney, S. H. Alageel, and E. L. Godfrey. 2019. "Behaviour Change Techniques Associated with Adherence to Prescribed Exercise in Patients with Persistent Musculoskeletal Pain: Systematic Review." *British Journal of Health Psychology* 24 (1): 10–30. https://doi.org/10.1111/bjhp.12324.

Meng, Q., and J. Kang. 2016. "Effect of Sound-Related Activities on Human Behaviours and Acoustic Comfort in Urban Open Spaces." *Science of the Total Environment* 573: 481–493. https://doi.org/10.1016/j.scitotenv.2016.08.130.

Mertens, S., M. Herberz, U. J. Hahnel, and T. Brosch. 2022. "The Effectiveness of Nudging: A Meta-Analysis of Choice Architecture Interventions Across Behavioral Domains." *Proceedings of the National Academy of Sciences* 119 (1): e2107346118. https://doi.org/10.1073/pnas.2107346118.

Michie, S., L. Atkins, and R. West. 2014. *The Behaviour Change Wheel. A Guide to Designing Interventions*. 1st ed. Great Britain: Silverback Publishing.

Michie, S., R. N. Carey, M. Johnston, A. J. Rothman, M. De Bruin, M. P. Kelly, and L. E. Connell. 2018. "From Theory-Inspired to Theory-Based Interventions: A Protocol for Developing and Testing a Methodology for Linking Behaviour Change Techniques to Theoretical Mechanisms of Action." *Annals of Behavioral Medicine* 52 (6): 501–512. https://doi.org/10.1007/s12160-016-9816-6.

Michie, S., M. Richardson, M. Johnston, C. Abraham, J. Francis, W. Hardeman, M. P. Eccles, J. Cane, and C. E. Wood. 2013. "The Behavior Change Technique Taxonomy (v1) of 93 Hierarchically Clustered Techniques: Building an International Consensus for the Reporting of Behavior Change Interventions." *Annals of Behavioral Medicine* 46 (1): 81–95. http://dx.doi.org/10.1007/s12160-013-9486-6.

Michlewski, K. 2008. "Uncovering Design Attitude: Inside the Culture of Designers." *Organization Studies* 29 (3): 373–392. https://doi.org/10.1177/0170840607088019.

Milkman, K. L., et al. 2021. "Megastudies Improve the Impact of Applied Behavioural Science." *Nature* 600: 478–483. https://doi.org/10.1038/s41586-021-04128-4.

Miltenberger, R. G. 2016. *Behavior Modification: Principles and Procedures*. Boston, MA: Cengage Learning.

Mirosa, M., H. Munro, E. Mangan-Walker, and D. Pearson. 2016. "Reducing Waste of Food Left on Plates: Interventions Based on Means-End Chain Analysis of Customers in Foodservice Sector." *British Food Journal* 118 (9): 2326–2343. https://doi.org/10.1108/BFJ-12-2015-0460.

Müller-Riemenschneider, F., T. Reinhold, M. Nocon, and S. N. Willich. 2008. "Long-term Effectiveness of Interventions Promoting Physical Activity: A Systematic Review." *Preventive Medicine* 47 (4): 354–368. https://doi.org/10.1016/j.ypmed.2008.07.006.

Nastasi, B. K., K. Varjas, S. L. Schensul, K. T. Silva, J. J. Schensul, and P. Ratnayake. 2000. "The Participatory Intervention Model: A Framework for Conceptualizing and Promoting Intervention Acceptability." *School Psychology Quarterly* 15 (2): 207–232. https://doi.org/10.1037/h0088785.

Newton, J. T., and K. Asimakopoulou. 2018. "Behavioral models for periodontal health and disease." *Periodontology 2000* 78 (1): 201–211. https://doi.org/10.1111/prd.12236.

Niedderer, K., S. Clune, and G. Ludden, eds. 2017. *Design for Behaviour Change: Theories and Practices of Designing for Change*. Oxfordshire: Routledge.
Nielsen, C. K. B. B., J. Daalhuizen, and P. J. Cash. 2021. "Defining the Behavioural Design Space." *International Journal of Design* 15 (1): 1–16.
Nogueira, A., and R. Schmidt. 2022. "Participatory Policy Design: Igniting Systems Change Through Prototyping." *Policy Design and Practice* 5 (1): 32–50. https://doi.org/10.1080/25741292.2021.1888399.
Norman, D., J. Miller, and A. Henderson. 1995. "What you see, Some of What's in the Future, and how we go About Doing it: Hi at Apple Computer." In *Conference Companion on Human Factors in Computing Systems* (p. 155).
Northridge, M. E., and S. S. Metcalf. 2016. "Enhancing Implementation Science by Applying Best Principles of Systems Science." *Health Research Policy and Systems* 14 (1): 1–8. https://doi.org/10.1186/s12961-016-0146-8.
Ohno, T. 1988. *Toyota Production System: Beyond Large-Scale Production*. Boca Raton, FL: CRC Press.
Osborn, A. F. 1953. *Applied Imagination: Principles and Procedures of Creative Thinking*. New York City: Scribners.
Page, Matthew J., Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, Larissa Shamseer, et al. 2021. "The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews." *International Journal of Surgery* 88: 105906. https://doi.org/10.1016/j.ijsu.2021.105906.
Palsola, M., E. Renko, K. Kostamo, F. Lorencatto, and N. Hankonen. 2020. "Thematic Analysis of Acceptability and Fidelity of Engagement for Behaviour Change Interventions: The Let's Move It Intervention Interview Study." *British Journal of Health Psychology* 25 (3): 772–789. https://doi.org/10.1111/bjhp.12433.
Panter, J., C. Guell, R. Prins, and D. Ogilvie. 2017. "Physical Activity and the Environment: Conceptual Review and Framework for Intervention Research." *International Journal of Behavioral Nutrition and Physical Activity* 14 (1): 1–13. https://doi.org/10.1186/s12966-017-0610-z.
Patton, D. E., C. Ryan, and C. M. Hughes. 2020. "Development of a Complex Community Pharmacy Intervention Package Using Theory-Based Behaviour Change Techniques to Improve Older Adults' Medication Adherence." *BMC Health Services Research* 20 (1): 1–12. https://doi.org/10.1186/s12913-020-05282-7.
Peeters, A. L., E. van der Werff, and N. Tromp. 2022. "Designing for Value-Behaviour Consistency: Ethical Choice Architecture to Stimulate Sustainable Meat Purchase." *Cleaner and Responsible Consumption* 5: 100067. https://doi.org/10.1016/j.clrc.2022.100067.
Perski, O., D. Szinay, E. Corker, L. Shahab, R. West, and S. Michie. 2022. "Interventions to Increase Personal Protective Behaviours to Limit the Spread of Respiratory Viruses: A Rapid Evidence Review and Meta-Analysis." *British Journal of Health Psychology* 27 (1): 215–264. https://doi.org/10.1111/bjhp.12542.
Petrillo, A., F. De Felice, E. Jannelli, C. Autorino, M. Minutillo, and A. L. Lavadera. 2016. "Life Cycle Assessment (LCA) and Life Cycle Cost (LCC) Analysis Model for a Stand-Alone Hybrid Renewable Energy System." *Renewable Energy* 95: 337–355. https://doi.org/10.1016/j.renene.2016.04.027.
Pincus, T., S. Anwar, L. M. McCracken, A. McGregor, L. Graham, M. Collinson, J. McBeth, et al. 2015. "Delivering an Optimised Behavioural Intervention (OBI) to people with low back pain with high psychological risk; results and lessons learnt from a feasibility randomised controlled trial of Contextual Cognitive Behavioural Therapy (CCBT) vs. Physiotherapy." *BMC Musculoskeletal Disorders* 16 (1): 1–11. http://dx.doi.org/10.1186/s12891-015-0594-2.
Preece, S. J., N. Brookes, A. E. Williams, R. K. Jones, C. Starbuck, A. Jones, and N. E. Walsh. 2021. "A new Integrated Behavioural Intervention for Knee Osteoarthritis: Development and Pilot Study." *BMC Musculoskeletal Disorders* 22 (1): 1–14. https://doi.org/10.1186/s12891-021-04389-0.
Presseau, J., N. McCleary, F. Lorencatto, A. M. Patey, J. M. Grimshaw, and J. J. Francis. 2019. "Action, Actor, Context, Target, Time (AACTT): A Framework for Specifying Behaviour." *Implementation Science* 14 (1): 1–13. https://doi.org/10.1186/s13012-019-0951-x.
Reid, S., and R. Schmidt. 2018. "A New Model for Integrating Behavioral Science and Design." *Behavioral Scientist*.

Reinertsen, D. G. 1985. "Blitzkrieg Product Development: Cut Development in Half: Electronic Business." *January* 15.

Renzi, C., F. Leali, and L. Di Angelo. 2017. "A Review on Decision-Making Methods in Engineering Design for the Automotive Industry." *Journal of Engineering Design* 28 (2): 118–143. https://doi.org/10.1080/09544828.2016.1274720.

Reynolds, J. P., S. Archer, M. Pilling, M. Kenny, G. J. Hollands, and T. M. Marteau. 2019. "Public Acceptability of Nudging and Taxing to Reduce Consumption of Alcohol, Tobacco, and Food: A Population-Based Survey Experiment." *Social Science and Medicine* 236: 112395. https://doi.org/10.1016/j.socscimed.2019.112395.

Rodrigues, A. D. S. L., L. B. A. Martinez, and Z. C. Silveira. 2024. "An Iterative Design Procedure for the Development of Assistive Devices Based on a Participatory Approach." *Journal of the Brazilian Society of Mechanical Sciences and Engineering* 46 (3): 1–21.

Roozenburg, N. F. 1993. "On the Pattern of Reasoning in Innovative Design." *Design Studies* 14 (1): 4–18. https://doi.org/10.1016/S0142-694X(05)80002-X.

Rosenberger, E. M., A. J. DeVito Dabbs, A. F. DiMartini, D. P. Landsittel, J. M. Pilewski, and M. A. Dew. 2017. "Long-Term Follow-up of a Randomized Controlled Trial Evaluating a Mobile Health Intervention for Self-Management in Lung Transplant Recipients." *American Journal of Transplantation* 17 (5): 1286–1293. https://doi.org/10.1111/ajt.14062.

Sääksjärvi, M., and M. Gonçalves. 2018. "Creativity and Meaning: Including Meaning as a Component of Creative Solutions." *Ai Edam* 32 (4): 365–379.

Sanders, M., V. Snijders, and M. Hallsworth. 2018. "Behavioural Science and Policy: Where are we now and Where are we Going?" *Behavioural Public Policy* 2 (2): 144–167. https://doi.org/10.1017/bpp.2018.17.

Schmidt, R. 2022. "A Model for Choice Infrastructure: Looking Beyond Choice Architecture in Behavioral Public Policy." *Behavioural Public Policy*, 1–26.

Schmidt, R., and K. Stenger. 2021. "Behavioral Brittleness: The Case for Strategic Behavioral Public Policy." *Behavioural Public Policy*, 1–26. https://doi.org/10.1017/bpp.2021.16.

Schroé, H., C. Van Der Mispel, I. De Bourdeaudhuij, M. Verloigne, L. Poppe, and G. Crombez. 2019. "A Factorial Randomised Controlled Trial to Identify Efficacious Self-Regulation Techniques in an e- and m-Health Intervention to Target an Active Lifestyle: Study Protocol." *Trials* 20 (1): 1–14. https://doi.org/10.1186/s13063-019-3456-7.

Schroé, H., D. Van Dyck, A. De Paepe, L. Poppe, W. W. Loh, M. Verloigne, … G. Crombez. 2020. "Which Behaviour Change Techniques are Effective to Promote Physical Activity and Reduce Sedentary Behaviour in Adults: A Factorial Randomized Trial of an e- and m-Health Intervention." *International Journal of Behavioral Nutrition and Physical Activity* 17 (1): 1–16. https://doi.org/10.1186/s12966-020-01001-x.

Sekhon, M., M. Cartwright, and J. J. Francis. 2017. "Acceptability of Healthcare Interventions: An Overview of Reviews and Development of a Theoretical Framework." *BMC Health Services Research* 17 (1): 1–13. https://doi.org/10.1186/s12913-017-2031-8.

Shahsavari, H., P. Matourypour, S. Ghiyasvandian, and M. R. G. Nejad. 2020. "Medical Research Council Framework for Development and Evaluation of Complex Interventions: A Comprehensive Guidance." *Journal of Education and Health Promotion* 9 (1): 88. https://doi.org/10.4103/jehp.jehp_649_19.

Skivington, K., L. Matthews, S. A. Simpson, P. Craig, J. Baird, J. M. Blazeby, … L. Moore. 2021. "A New Framework for Developing and Evaluating Complex Interventions: Update of Medical Research Council Guidance." *bmj* 374.

Smith, S. A., A. A. Lake, C. Summerbell, V. Araujo-Soares, and F. Hillier-Brown. 2016. "The Effectiveness of Workplace Dietary Interventions: Protocol for a Systematic Review and Meta-Analysis." *Systematic Reviews* 5 (1): 1–6. https://doi.org/10.1186/s13643-016-0200-1.

Smith, P. G., and D. G. Reinertsen. 1998. *Developing Products in Half the Time: New Rules, new Tools*. New York: Van Nostrand Reinhold.

Stavropoulos, P., C. Giannoulis, A. Papacharalampopoulos, P. Foteinopoulos, and G. Chryssolouris. 2016. "Life Cycle Analysis: Comparison Between Different Methods and Optimization Challenges." *Procedia CIRP* 41: 626–631. https://doi.org/10.1016/j.procir.2015.12.048.

Stegenga, J. 2015. "Measuring Effectiveness. Studies in History and Philosophy of Science Part C." *Studies in History and Philosophy of Biological and Biomedical Sciences* 54: 62–71. https://doi.org/10.1016/j.shpsc.2015.06.003.

Stingl, V., and J. Geraldi. 2017. "Errors, Lies and Misunderstandings: Systematic Review on Behavioural Decision Making in Projects." *International Journal of Project Management* 35 (2): 121–135. https://doi.org/10.1016/j.ijproman.2016.10.009.

Sy, D. 2007. "Adapting Usability Investigations for Agile User-Centered Design." *Journal of Usability Studies* 2 (3): 112–132.

Szaszi, B., A. Higney, A. Charlton, A. Gelman, I. Ziano, B. Aczel, … E. Tipton. 2022. "No Reason to Expect Large and Consistent Effects of Nudge Interventions." *Proceedings of the National Academy of Sciences* 119 (31): e2200732119. https://doi.org/10.1073/pnas.2200732119.

Szaszi, B., A. Palinkas, B. Palfi, A. Szollosi, and B. Aczel. 2018. "A Systematic Scoping Review of the Choice Architecture Movement: Toward Understanding When and why Nudges Work." *Journal of Behavioral Decision Making* 31: 355–366. https://doi.org/10.1002/bdm.2035.

Tanna, Nuttan K, Emma C Alexander, Charlotte Lee, Monica Lakhanpaul, Rickin M Popat, Pamela Almeida-Meza, Alice Tuck, Logan Manikam, and Mitch Blair. 2021. "Interventions to Improve Vitamin D Status in at-Risk Ethnic Groups During Pregnancy and Early Childhood: A Systematic Review." *Public Health Nutrition* 24 (11): 3498–3519. https://doi.org/10.1017/S1368980021000756.

Tengland, P. A. 2012. "Behavior Change or Empowerment: On the Ethics of Health-Promotion Strategies." *Public Health Ethics* 5 (2): 140–153. https://doi.org/10.1093/phe/phs022.

Tesfaye, W. H., D. Erku, A. Mekonnen, Y. G. Tefera, R. Castelino, K. Sud, … K. Obamiro. 2021. "Medication non-adherence in chronic kidney disease: a mixed-methods review and synthesis using the theoretical domains framework and the behavioural change wheel." *Journal of Nephrology* 34 (4): 1091–1125. https://doi.org/10.1007/s40620-020-00895-x.

Thomas-Walters, L., S. Vieira, V. Jiménez, D. Monteiro, B. Ferreira, R. J. Smith, and D. Veríssimo. 2020. "Challenges in the Impact Evaluation of Behaviour Change Interventions: The Case of sea Turtle Meat and Eggs in São Tomé." *People and Nature* 2 (4): 913–922. https://doi.org/10.1002/pan3.10162.

Tolan, P. H., D. Gorman-Smith, D. Henry, and M. Schoeny. 2009. "The Benefits of Booster Interventions: Evidence from a Family-Focused Prevention Program." *Prevention Science* 10: 287–297. https://doi.org/10.1007/s11121-009-0139-8.

Tromp, N., and P. Hekkert. 2017. "The Hidden Influence of Design." In *Design for Behaviour Change*, edited by Kristina Niedderer, Stephen Clune, and Geke Ludden, 138–149. London: Routledge.

Tromp, N., and P. Hekkert. 2018. *Designing for Society: Products and Services for a Better World*. London: Bloomsbury Publishing.

Tromp, N., P. Hekkert, and P. P. Verbeek. 2011. "Design for Socially Responsible Behavior: A Classification of Influence Based on Intended User Experience." *Design Issues* 27 (3): 3–19. https://doi.org/10.1162/DESI_a_00087.

Tversky, A., and D. Kahneman. 1974. "Judgment Under Uncertainty: Heuristics and Biases." *Science* 185: 1124–1131. https://doi.org/10.1126/science.185.4157.1124.

Van Arkel, T., and N. Tromp. 2022. Designing Appropriate Things: An Experiential Perspective on the Effectiveness of Artefacts in Contributing to Behaviour Change. DRS2022: Bilbao, 25.

Van Boeijen, A., J. Daalhuizen, and J. Zijlstra. 2020. *Delft Design Guide: Perspectives, Models, Approaches, Methods*. Amsterdam. The Netherlands: BIS Publishers.

Van Buul, L. W., J. J. Sikkens, M. A. Van Agtmael, M. H. H. Kramer, J. T. Van der Steen, and C. M. P. M. Hertogh. 2014. "Participatory Action Research in Antimicrobial Stewardship: A Novel Approach to Improving Antimicrobial Prescribing in Hospitals and Long-Term Care Facilities." *Journal of Antimicrobial Chemotherapy* 69 (7): 1734–1741. https://doi.org/10.1093/jac/dku068.

Van den Hoven, J., P. E. Vermaas, and I. Van de Poel. 2015. *Handbook of Ethics, Values, and Technological Design: Sources, Theory, Values and Application Domains*. Springer.

Van Halen, Cees, Carlo Vezzoli, and Robert Wimmer. 2005. *Methodology for Product Service System Innovation*. Assen: Uitgeverij Van Gorcum, 21.

Vargas-Garcia, E. J., C. E. L. Evans, and J. E. Cade. 2015. "Impact of Interventions to Reduce Sugar-Sweetened Beverage Intake in Children and Adults: A Protocol for a Systematic Review and Meta-Analysis." *Systematic Reviews* 4 (1): 1–8. https://doi.org/10.1186/2046-4053-4-1.

Vasantha, G. V. A., R. Roy, A. Lelah, and D. Brissaud. 2012. "A Review of Product–Service Systems Design Methodologies." *Journal of Engineering Design* 23 (9): 635–659. https://doi.org/10.1080/09544828.2011.639712.

Voorheis, Paula, Albert Zhao, Kerry Kuluski, Quynh Pham, Ted Scott, Peter Sztur, Nityan Khanna, Mohamed Ibrahim, and Jeremy Petch. 2022. "Integrating Behavioral Science and Design Thinking to Develop Mobile Health Interventions: Systematic Scoping Review." *JMIR MHealth and UHealth* 10 (3): e35799. https://doi.org/10.2196/35799.

Waddock, S., G. M. Meszoely, S. Waddell, and D. Dentoni. 2015. "The Complexity of Wicked Problems in Large Scale Change." *Journal of Organizational Change Management* 28 (6): 993–1012. https://doi.org/10.1108/JOCM-08-2014-0146.

Wang, S., and C. K. Jiao. 2022. "Leveraging Behavioural Economics in Smart Nudge Design Through Data-Driven Prospect-Theoretic Modelling and Context-Aware Intelligent Reasoning: Application to Smart tip Nudging." *Journal of Engineering Design* 33 (11): 896–918. https://doi.org/10.1080/09544828.2022.2150448.

Wee, S. L., and H. J. M. Vrijhoef. 2015. "A Conceptual Framework for Evaluating the Conceptualization, Implementation and Performance of Transitional Care Programmes." *Journal of Evaluation in Clinical Practice* 21 (2): 221–228. https://doi.org/10.1111/jep.12292.

Wendel, S. A. 2013. *Designing for Behavior Change*. Sebastopol, CA: O'Reilly.

Westland, H., I. D. Bos-Touwen, J. C. A. Trappenburg, C. D. Schröder, N. J. de Wit, and M. J. Schuurmans. 2017. "Unravelling Effectiveness of a Nurse-led Behaviour Change Intervention to Enhance Physical Activity in Patients at Risk for Cardiovascular Disease in Primary Care: Study Protocol for a Cluster Randomised Controlled Trial." *Trials* 18 (1): 1–15. https://doi.org/10.1186/s13063-017-1823-9.

Wilson, A. L., E. Buckley, J. D. Buckley, and S. Bogomolova. 2016. "Nudging Healthier Food and Beverage Choices Through Salience and Priming. Evidence From a Systematic Review." *Food Quality and Preference* 51: 47–64. https://doi.org/10.1016/j.foodqual.2016.02.009.

Appendix

Table A1. Overview of papers in sample.

No.	Paper type	Author	Year	Journal	Database
P1	Intervention study	Palsola et al.	2020	British Journal of Health Psychology	Scopus
P2	Intervention study	Band et al.	2017	Implementation Science	PubMed
P3	Intervention study	Bowers et al.	2020	BMJ open	PubMed
P4	Intervention study	Knight et al.	2021	Public health	PubMed
P5	Intervention study	Costello et al.	2018	International journal of sport nutrition and exercise metabolism	Scopus
P6	Intervention study	Thomas-Walters et al.	2020	People and Nature	Scopus
P7	Intervention study	Patton et al.	2020	BMC Health Services Research	Scopus
P8	Intervention study	Preece et al.	2021	BMC Musculoskeletal Disorders	PubMed
P9	Intervention study	Harries & Rettie	2016	Sociology of Health & Illness	PubMed
P10	Intervention study	Pincus et al.	2015	BMC musculoskeletal disorders	PubMed
P11	Intervention study	Goyal et al.	2016	Canadian journal of diabetes	PubMed
P12	Intervention study	Liew et al.	2021	Physiotherapy	PubMed
P13	Intervention study	Barnett et al.	2014	Implementation Science	PubMed
P14	Intervention study	Schroé et al.	2020	International Journal of Behavioural Nutrition and Physical Activity	Scopus
P15	Review	Bentley et al.	2019	Behaviour Research and Therapy	WoS
P16	Review	Meade et al.	2019	British journal of health psychology	Scopus
P17	Review	der Wardt et al.	2021	British Journal of General Practice	Scopus
P18	Review	Dogru et al.	2021	Transportation research part F: traffic psychology and behaviour	Scopus
P19	Review	Langdridge et al.	2019	British Journal of Health Psychology	PubMed
P20	Review	Lambert et al.	2017	BMC public health	PubMed
P21	Review	Dyson & Cowdell	2021	International Journal for Quality in Health Care	PubMed
P22	Review	Lee et al.	2020	BMJ open	PubMed
P23	Review	Tanna et al.	2021	Public Health Nutrition	PubMed
P24	Review	Desveaux et al.	2019	Systematic reviews	PubMed
P25	Review	Beall et al.	2014	European journal of clinical investigation	PubMed
P26	Review	Panter et al.	2017	International Journal of Behavioural Nutrition and Physical Activity	PubMed
P27	Review	Perski et al.	2022	British Journal of Health Psychology	Scopus
P28	Review	Hulscher & Prins	2017	Clinical Microbiology and Infection	PubMed
P29	Review	Newton & Asimakopoulou	2018	Periodontology 2000	PubMed
P30	Review	Kassavou & Sutton	2018	Health Psychology Review	Scopus
P31	Review	Johnson & May	2015	BMJ open	Scopus
P32	Review	Tesfaye et al.	2021	Journal of Nephrology	PubMed
P33	Review	Atkins et al.	2020	Implementation Science	PubMed
P34	Protocol method	Miles et al.	2018	Trials	WoS
P35	Protocol method	Jovicic et al.	2015	Systematic reviews	Scopus

(*continued*).

Table A1. Continued.

No.	Paper type	Author	Year	Journal	Database
P36	Protocol method	Vargas-Garcia	2015	Systematic reviews	Scopus
P37	Protocol method	Smith et al.	2016	Systematic reviews	Scopus
P38	Protocol method	Bell et al.	2020	Sustainability	WoS
P39	Protocol method	Masrur et al.	2020	JBI evidence synthesis	Scopus
P40	Protocol method	Westland et al.	2017	Trials	Scopus
P41	Protocol method	Kaur et al.	2022	PloS one	PubMed
P42	Theory	Presseau et al.	2019	Implementation Science	WoS
P43	Theory	Heino et al.	2021	Behavioural Sciences	WoS
P44	Theory	Borek et al.	2015	BMC Public Health	Scopus
P45	Theory	Wee & Vrijhoef	2015	Journal of evaluation in clinical practice	Scopus
P46	Theory	Keyworth et al.	2020	British journal of health psychology	Scopus
P47	Theory	Andrade et al.	2020	BMJ open	Scopus
P48	Theory	Langley et al.	2021	Journal of Public Health	Scopus
P49	Theory	Reynolds et al.	2019	Social Science & Medicine	Scopus
P50	Theory	Mirosa et al.	2016	British Food Journal	Scopus
P51	Theory	Kok et al.	2016	Health psychology review	PubMed
P52	Theory	Northridge & Metcalf	2016	Health Research Policy and Systems	PubMed
P53	Theory	Klusmann et al.	2021	The Journals of Gerontology: Series B	PubMed
P54	Theory	Dunn et al.	2015	Health Technology Assessment	Scopus
P55	Theory	Van Buul et al.	2014	Journal of Antimicrobial Chemotherapy	PubMed
P56	Theory	Bondaronek et al.	2019	BMJ open	Scopus
P57	Theory	Govender et al.	2017	Dysphagia	Scopus